Flipsky ESC Tool V1.1

User Manual

Contents

1 Overview

Welcome to Flipsky ESC Tool !

This ESC Tool is developed specially for Flipsky FT series ESCs. To get started, you can use it to control the motor, do input settings, firmware upgrade, restore factory default setup and data analysis etc. respectively.

Please read the user guide carefully. Flipsky shall not be liable for any defects that are caused by neglect, misuse or mistreatment by the customer, including improper installation or testing, or for any products that have been altered or modified in any way by the customer. Moreover, Flipsky shall not be liable for any defects that result from the customers design, specifications or instructions for such products. Flipsky deserves the right to change products design, appearance, specifications etc.

2 Operation Requirements

2.1 Hardware Requirement

- FT Series ESCs
- Type-C USB cable

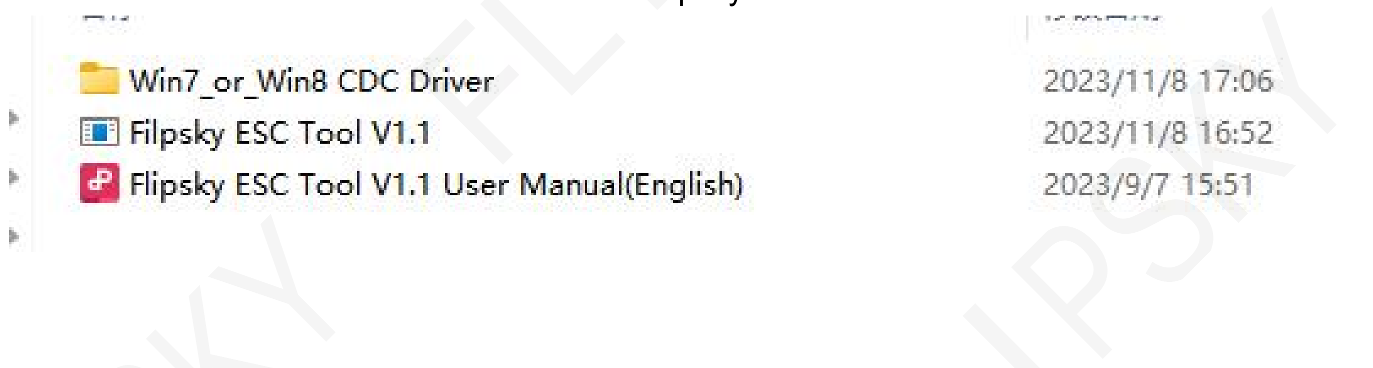
2.2 Computer System Requirement

- Windows 7 or above operating system is required.
- Install a CDC port driver when use Windows 7 or Windows 8 for ESC and ESC Tool recognition and connect.
- No need to install the driver for Windows 10 or above as the system built with it already.

2.2.1 System Driver Download and Related Files Description

Folder Name: Flipsky ESC Tool V1.1
Related files as below Screenshot 2-1.

Screenshot 2-1. Flipsky ESC Tool V1.1

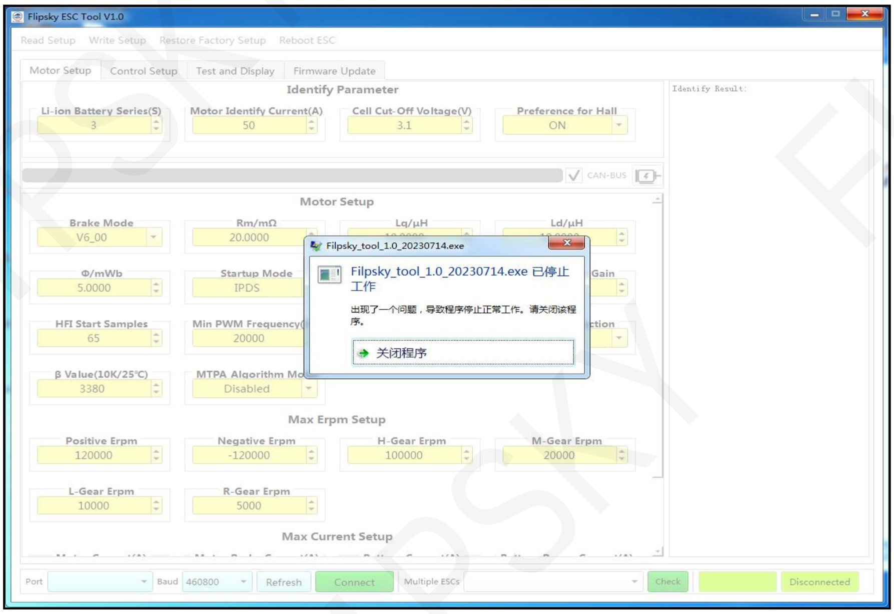


- Win7_or_ Win8 CDC Driver: Windows 7 / Windows 8 CDC port driver installation file
- Flipsky ESC Tool V1.1.exe
- Flipsky ESC Tool V1.1 User Manual

2.2.2 Driver Update / Reinstall for Windows 7 / Windows 8

Setting Windows 7 as an example:
The ESC cannot connect with the ESC Tool at the first time for no compatible system driver and show as below Screenshot 2-2. Please update / reinstall the port driver for your computer system to connect.

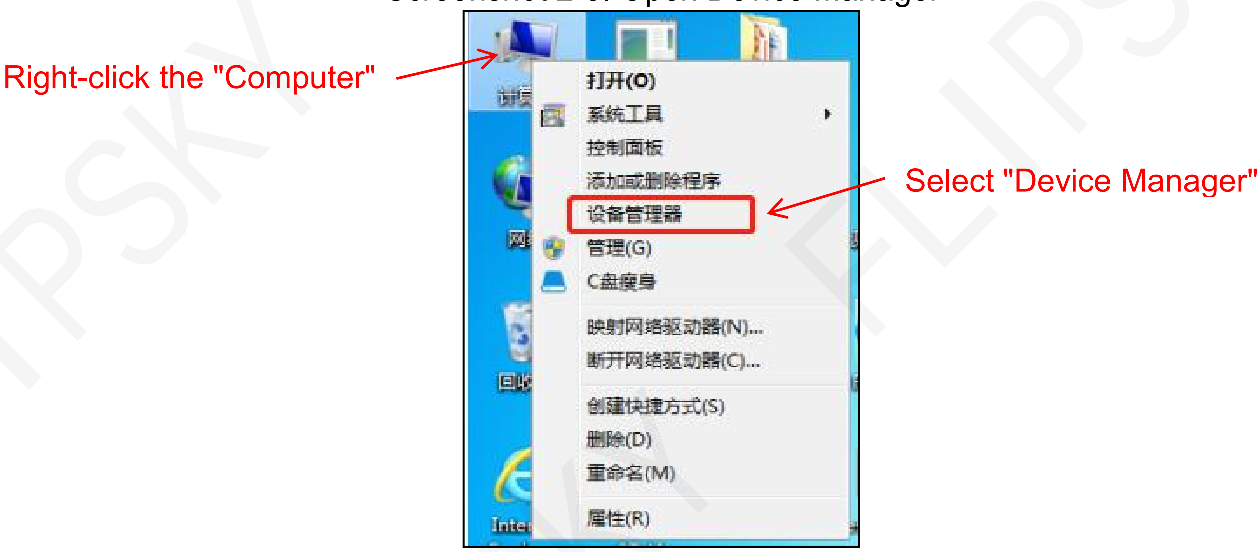
Screenshot 2-2. Fail to connect



Driver Update:

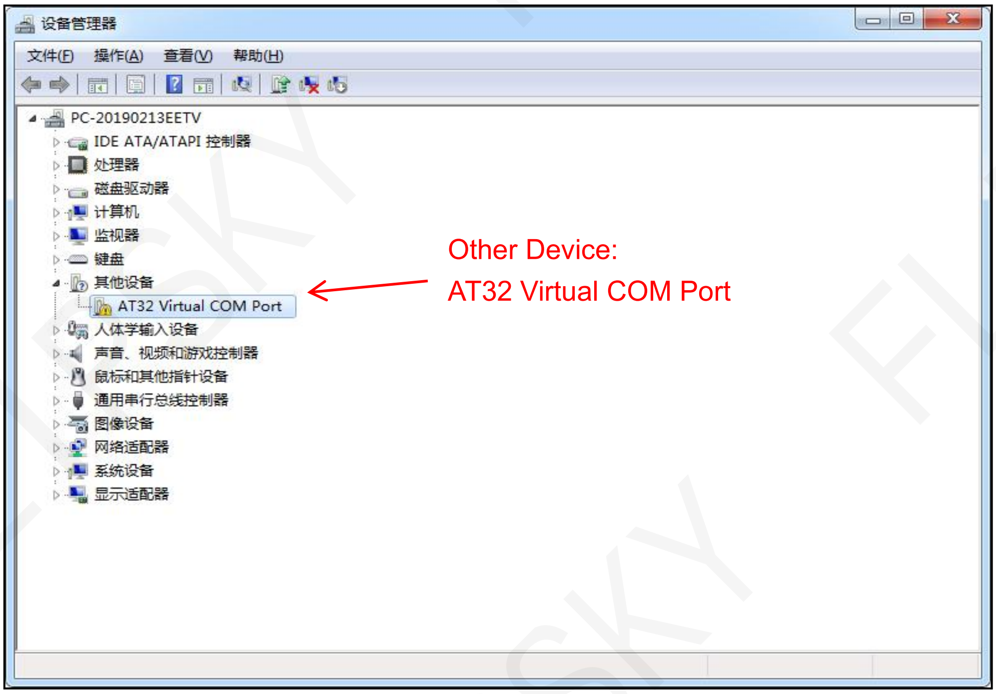
- Connect the ESC with the computer via the USB cable, right-click the "Computer" and select "Device Manager" as below Screenshot 2-3.

Screenshot 2-3. Open Device Manager



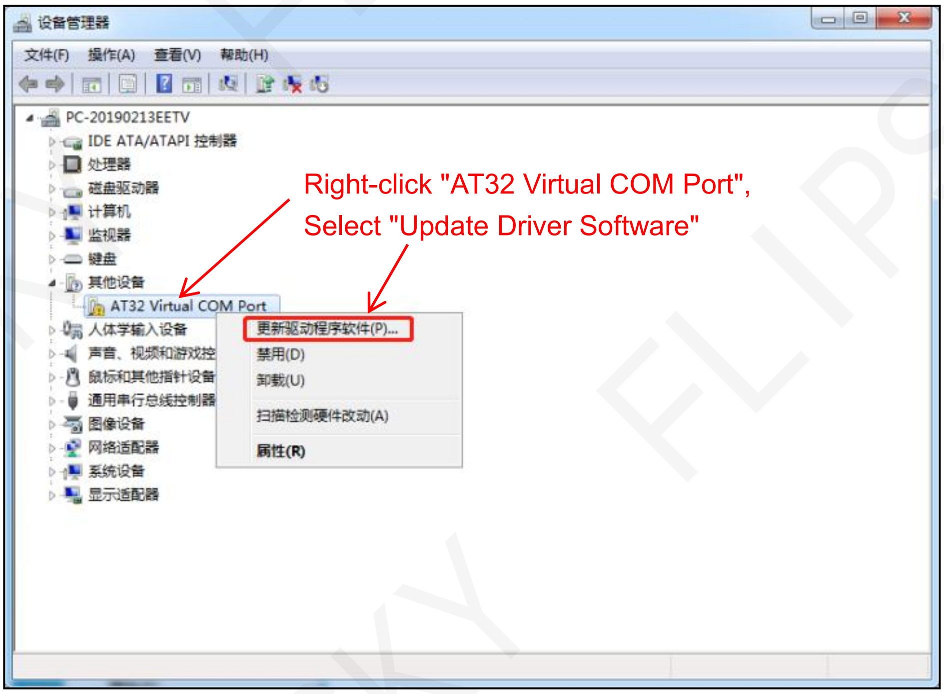
- When the "AT32 Virtual COM Port" with a yellow icon as below Screenshot 2-4, it means that you don't have the latest driver and it needs to be updated.

Screenshot 2-4. COM port driver cannot be used with a yellow icon



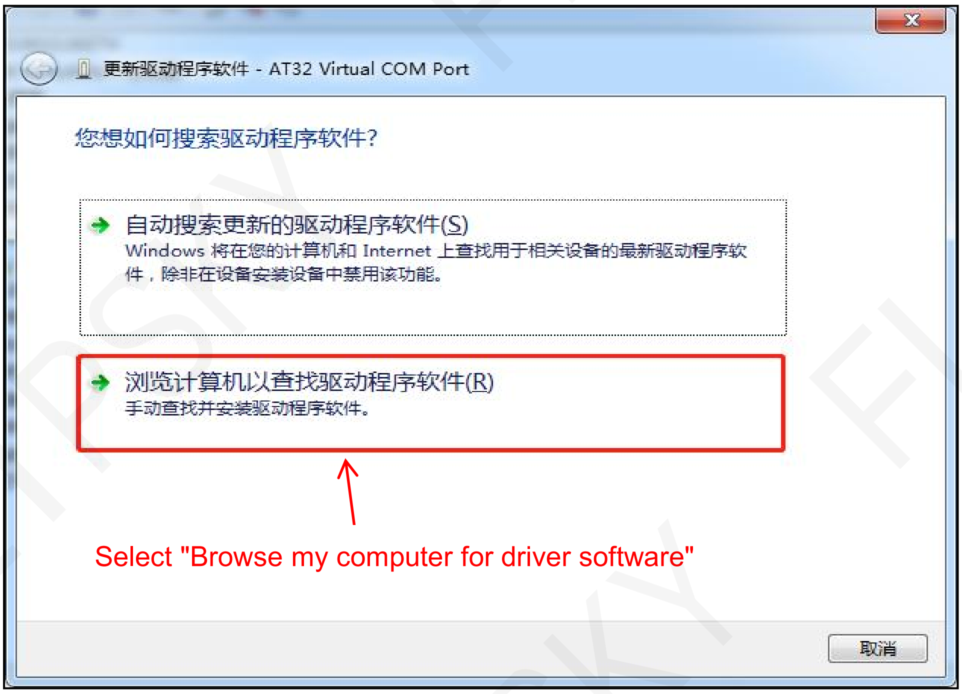
- Right-click the "AT32 Virtual COM Port" and select "Update Driver Software" as below Screenshot 2-5.

Screenshot 2-5. Select "Update Driver Software"



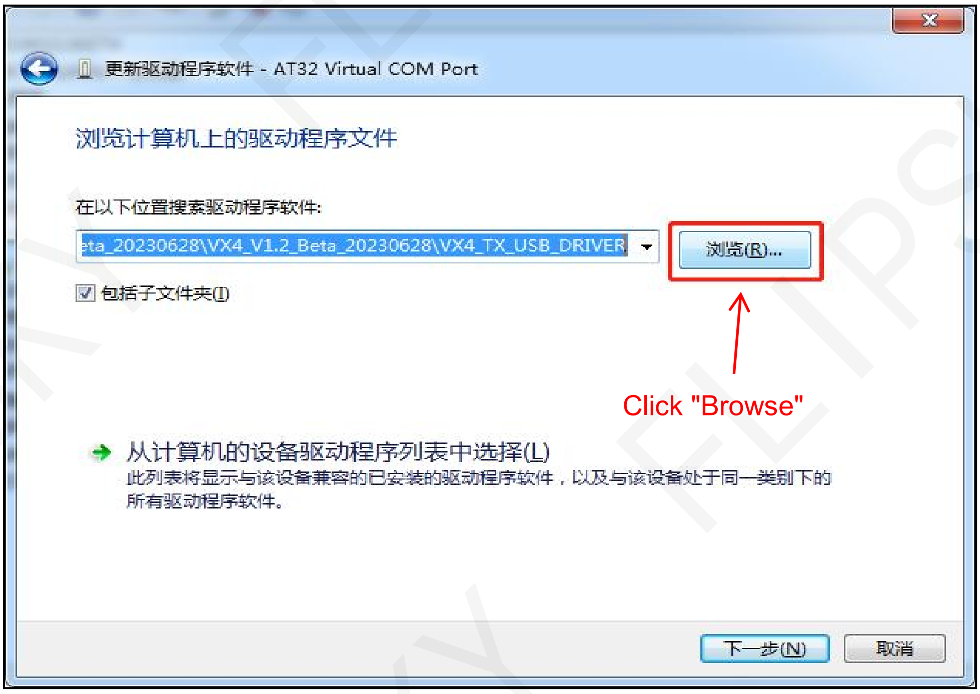
- Select "Browse my computer for driver software" as below Screenshot 2-6.

Screenshot 2-6. Select "Browse my computer for driver software"as below Screenshot 2-6



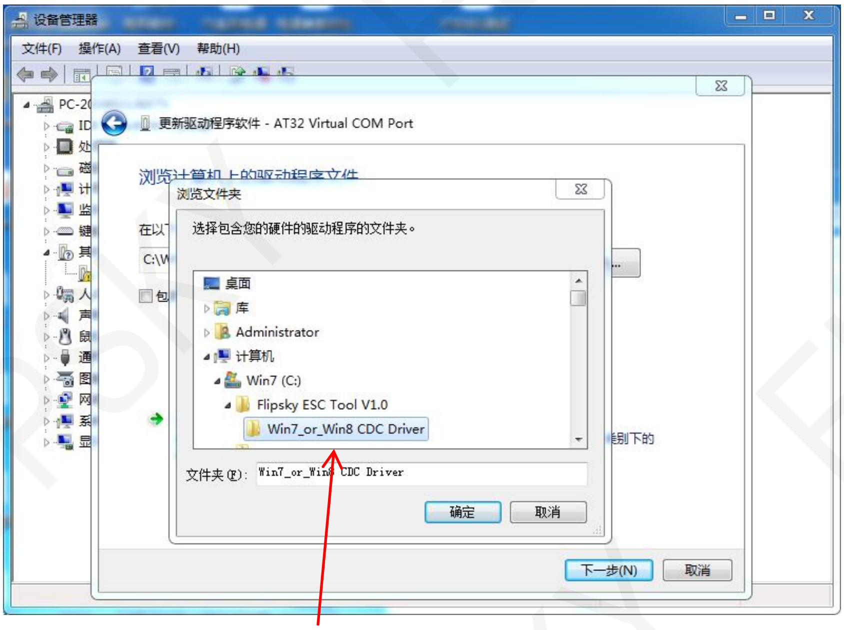
- Click "Browse" to find the driver file as below Screenshot 2-7.

Screenshot 2-7. Click "Browse" to find the location driver file



- Select the location driver installation folder as below example Screenshot 2-8.

Screenshot 2-8. Select the driver installation folder



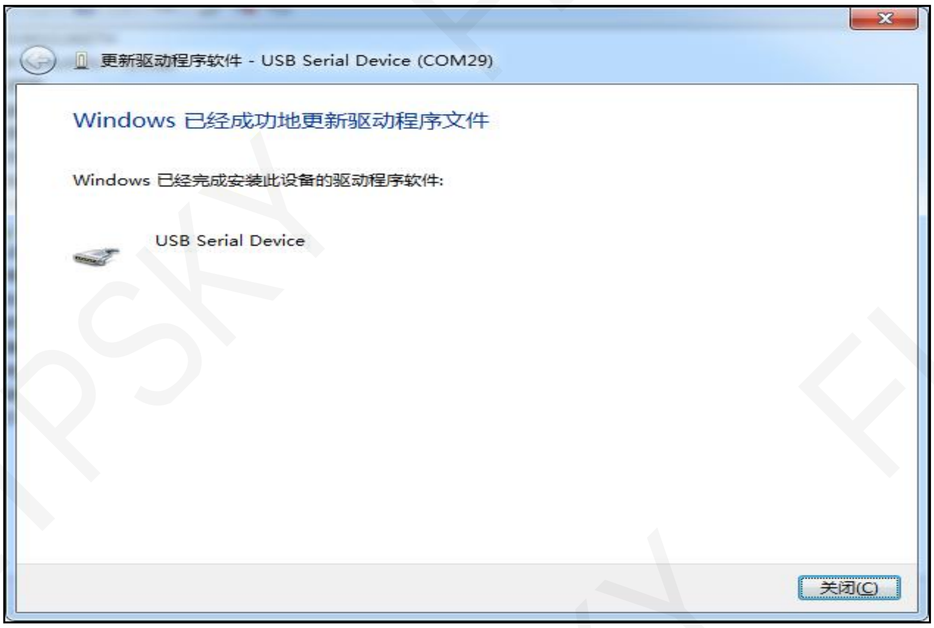
Find and Select"Win7_or_ Win8 CDC Driver"Folder,and Click “Next” .

- If a security prompt appears, select "Always install this driver software" as below Screenshot 2-9.

Screenshot 2-9. Select "Always install this driver software"

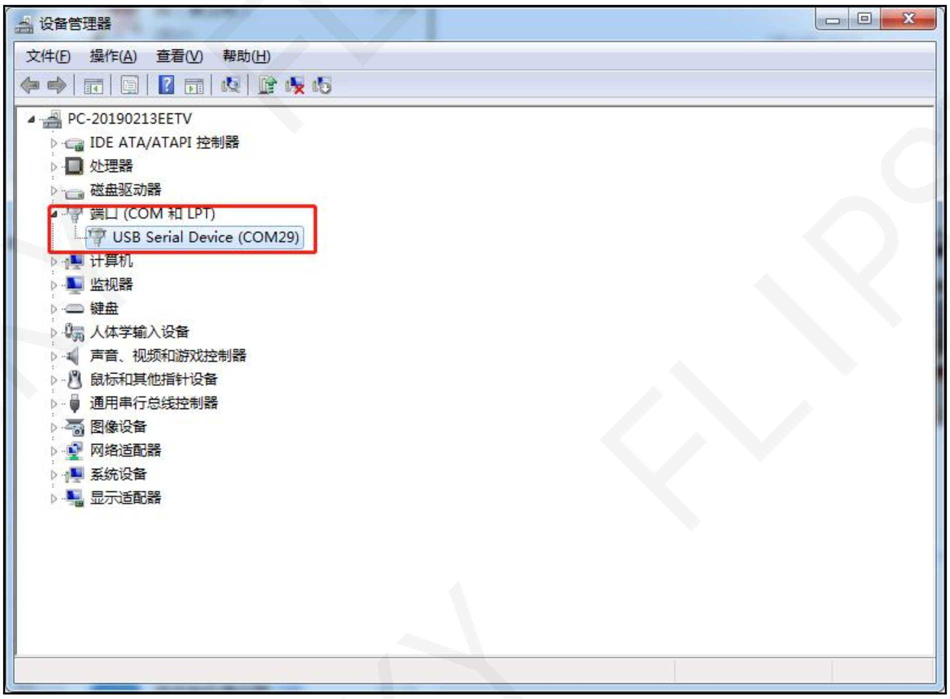
- And then, the driver update finish as below Screenshot 2-10.

Screenshot 2-10. The Driver Installation Finish



- Enter the "Device Manager" and check if no yellow icon before the USB Serial Device. If so, it means that the driver has been successfully updated as below Screenshot 2-11.

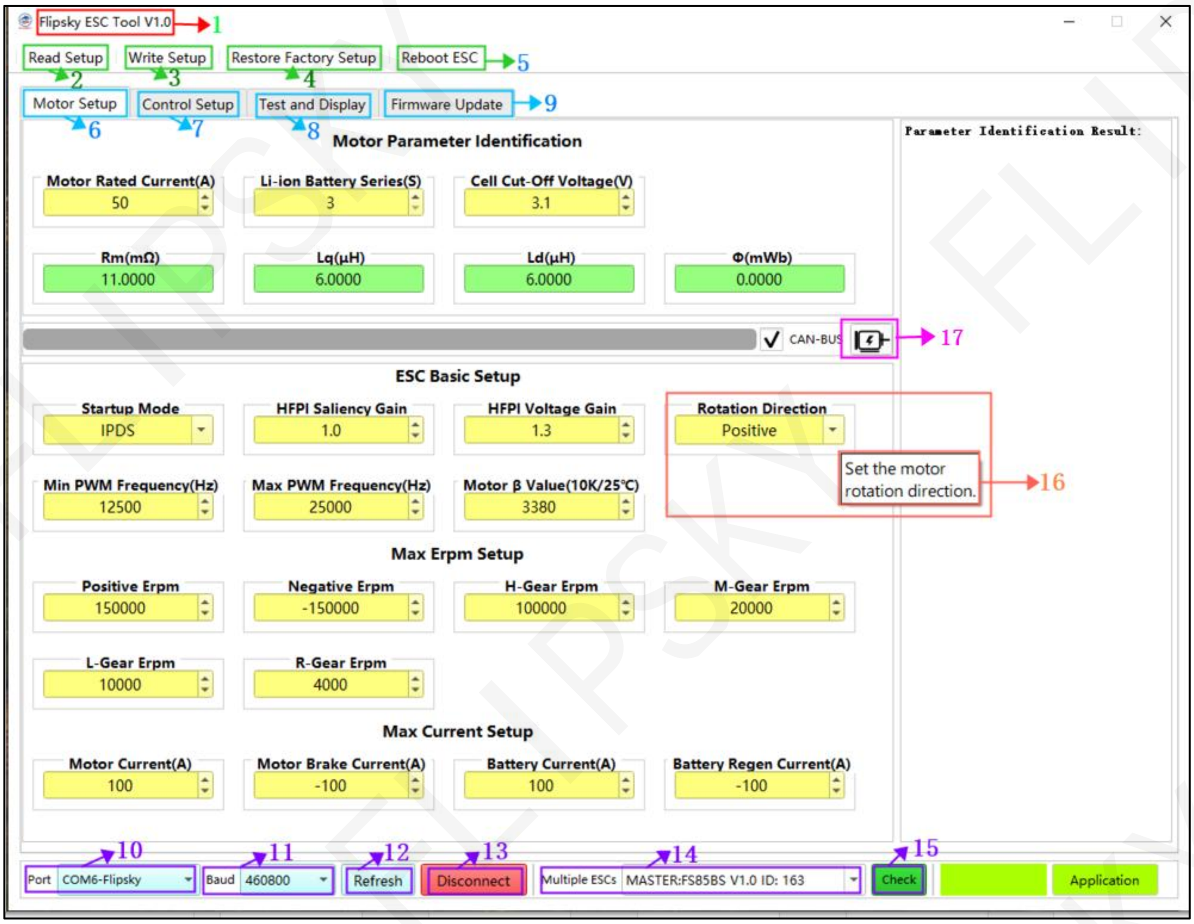
Screenshot 2-11. The USB Serial Device No Yellow Icon



3 Flipsky ESC Tool Home Interface Introduction

Flipsky ESC Tool home interface as below Screenshot 3-1.

Screenshot 3-1.Flipsky ESC Tool Home Interface



(1) Flipsky ESC Tool V1.1

(2) Read Setup: Read the current parameters of the ESC

(3) Write Setup: Write and save the current page parameters

(4) Restore Factory Setup: Restore factory default setup

(5) Reboot ESC: Click to re-connect the ESC

(6) Motor Setup: Click to start the motor FOC settings

(7) Control Setup: Click to start input control type setup

(8) Test and Display: Click to start the control test and data analysis

(9) Firmware Update: Click to start firmware bootloader and update

(10) Port: COM select

(11) Baud: Baud rate

(12) Refresh: Detection refresh port

(13) Connect / Disconnect: Connect or disconnect the ESC with the ESC Tool

(14) Multiple ESCs: CAN devices ID check

(15) Check: Scan all CAN devices

(16) Parameters Definitions: Explanation will be displayed when the computer mouse docked the parameters.

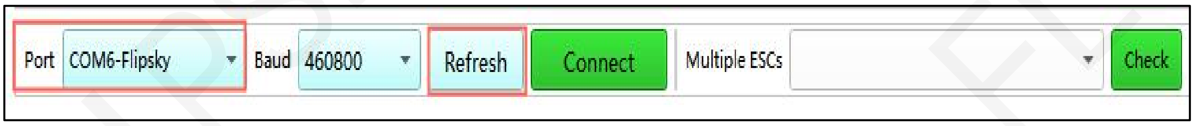
(17) Parameter identification result: click to start the parameters settings

4 Connection

4.1 FTESC and Flipsky ESC Tool Connection

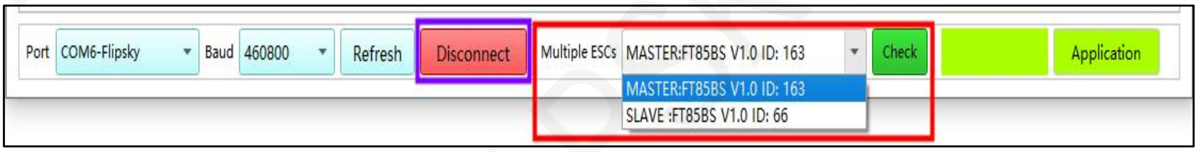
Power on the ESC and connect it with the computer via USB. Open the Flipsky ESC Tool V1.1 and the port for ESC ID (COM6-Flipsky) will be displayed. If no ID displays, please click "Refresh" to scan again as below Screenshot 4-1.

Screenshot 4-1. Port ESC ID Display and Refresh



Click "Connect" for ESC and ESC Tool connection. The ESC ID appears as below Screenshot 4-2.

Screenshot 4-2. ESC ID Display



The USB connection ESC is the master board, and all others are slave boards. Theoretically, 1 master board can be connected with a maximum of 253 slave CAN devices (except for the ID number for the master board, 0 and 255, others are fine). While, it is recommended not using more than 8 slave boards in practice since too many CAN devices affect the working performance efficiency. Moreover, please remove the CAN terminal resistors except for the first and last ESCs' CANs. Set 1pcs Master ESC and 6pcs Slave ESCs as an example:

Master(Start)==Slave1==Slave2==Slave3==Slave4==Slave5==Slave6(End)

Remove the CAN terminal resistors of Slave1, Slave2, Slave3, Slave4 and Slave5, leaving only the CAN terminal resistors of Master (Start) and Slave6 (End)

4.2 CAN Devices Connection Problem

If multiple ESCs are connected via CAN but the ESC Tool fails to scan all devices, 3 possible reasons for failure as belows:

1) The slave ESC boards' CAN terminal resistors haven't been removed.
2) Different ESCs with same IDs
3) One of ESCs ' CAN baud rate is inconsistent with others.

Troubleshooting:

1) Make sure that the ESCs' CAN terminal resistors have been removed correctly except for the

Start and the End CAN ESCs (section 4.1).

2) Unplug the CAN-BUS wires. Check the ESCs IDs one by one by connecting them with the ESC Tool separately via USB cable. If same IDs happen, revise them to make sure the uniqueness of the ESC ID.

3) If still not workable and missing the ESCs, please check the ESC's CAN baud rate and make sure the consistency.

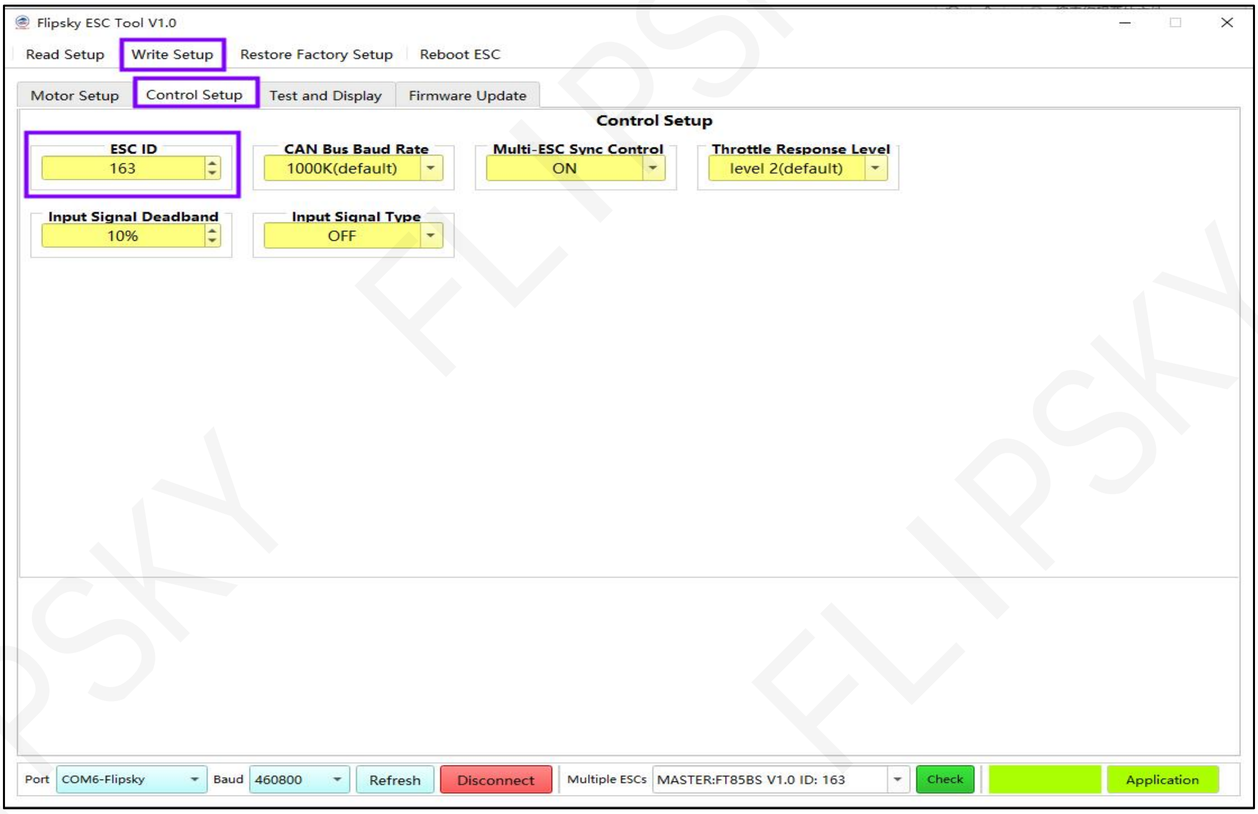
4) If still not succeed after the above trials, please restore the factory default setup and try to connect the ESCs again.

4.2.1 How to Set Multiple ESCs ID

The setting steps are shown as below Screenshot 4-3.

- Click “Control Setup” to enter the control setup interface
- Check the ESC ID and set the ID number from 1 to 254 (see details in section 7.1)
- Click the upper left menu "Write Setup" to save the setup.

Screenshot 4-3.ESC ID Setup

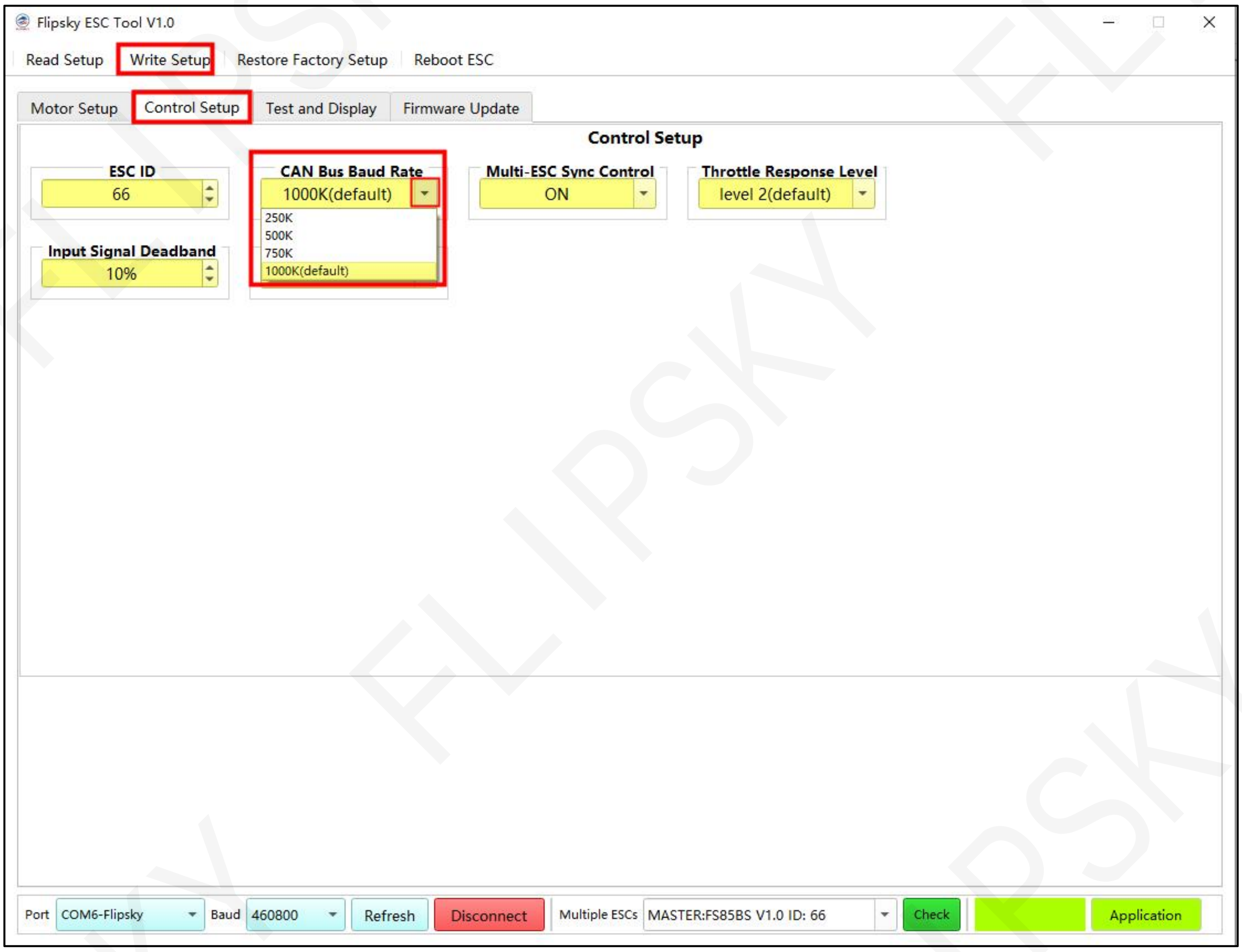


4.2.2 How to Set CAN Baud Rate

The setting steps are shown as below Screenshot 4-4.

- Click “Control Setup” to enter the control setup interface
- Check the CAN-Bus Baud Rate (see details in section 7.2)
- Click the upper left menu "Write Setup" to save the setup

Screenshot 4-4. CAN-Bus Baud Rate Setup

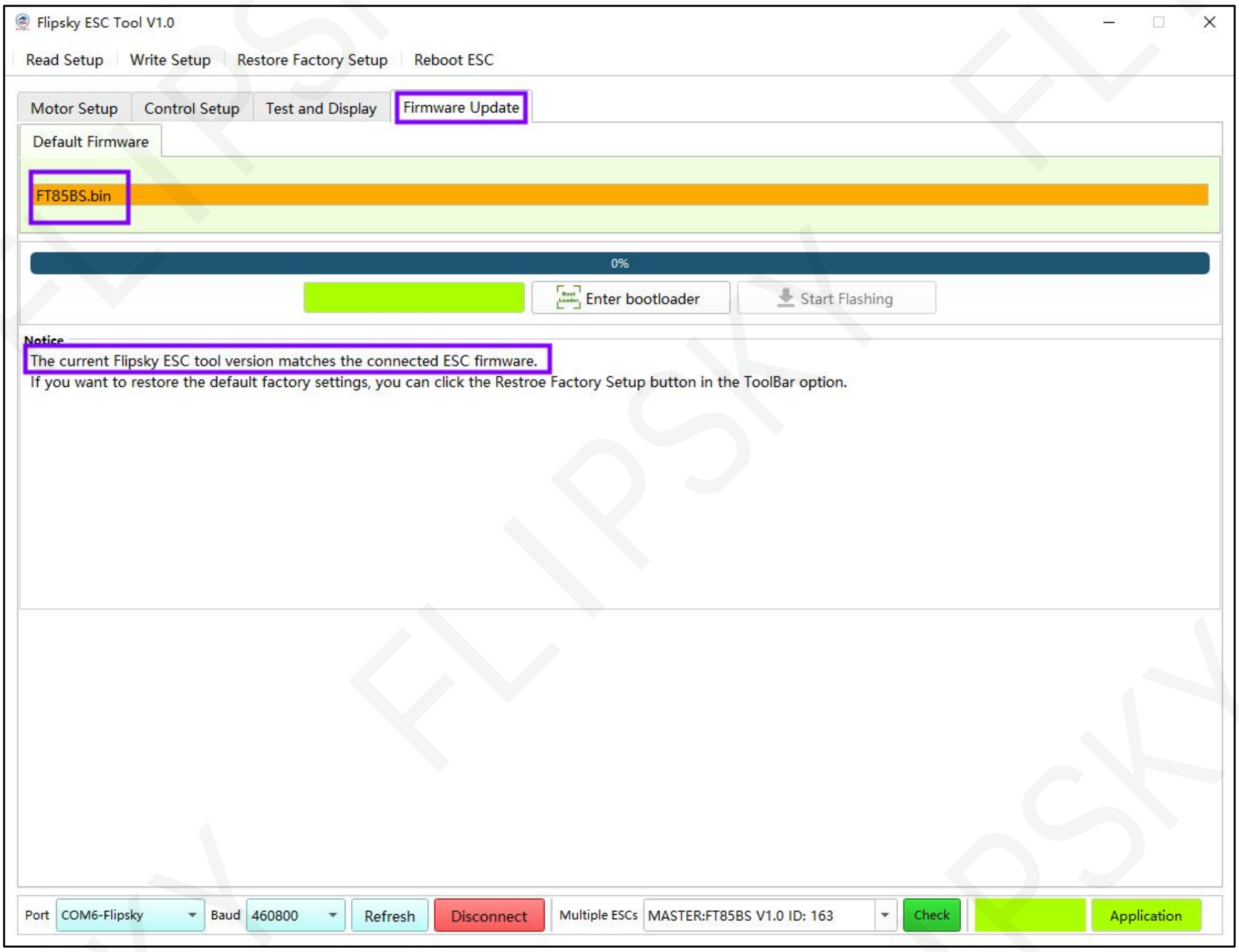


5 Firmware Update

5.1 ESC Firmware Version Check

Click "Firmware Update" to enter the firmware update interface, and it will check if the ESC firmware version matches the Flipsky ESC Tool version automatically. If not matching, please update the firmware as below Screenshot 5-1.

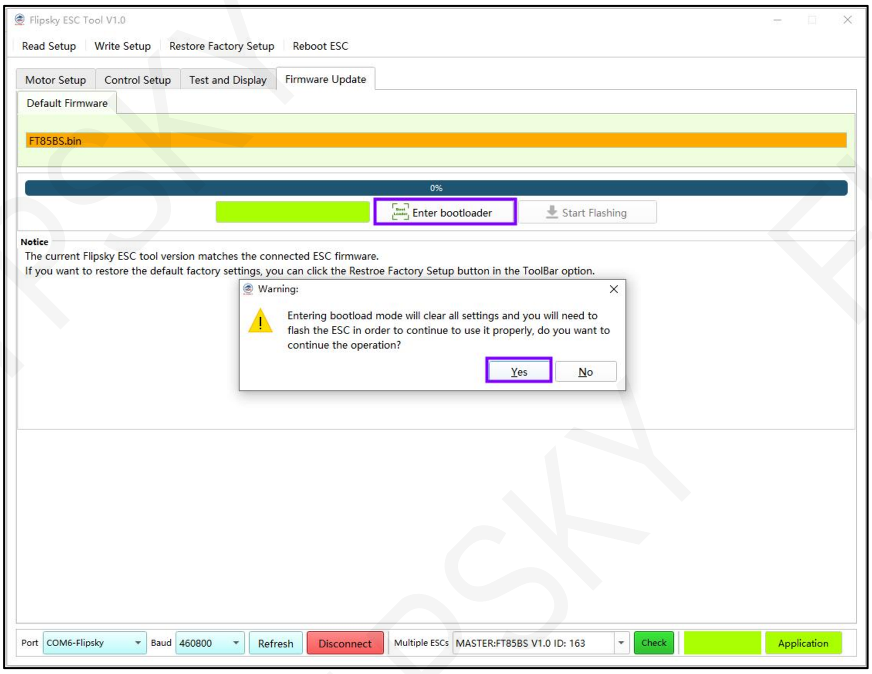
Screenshot 5-1 Firmware Update Interface



5.2 Firmware Update

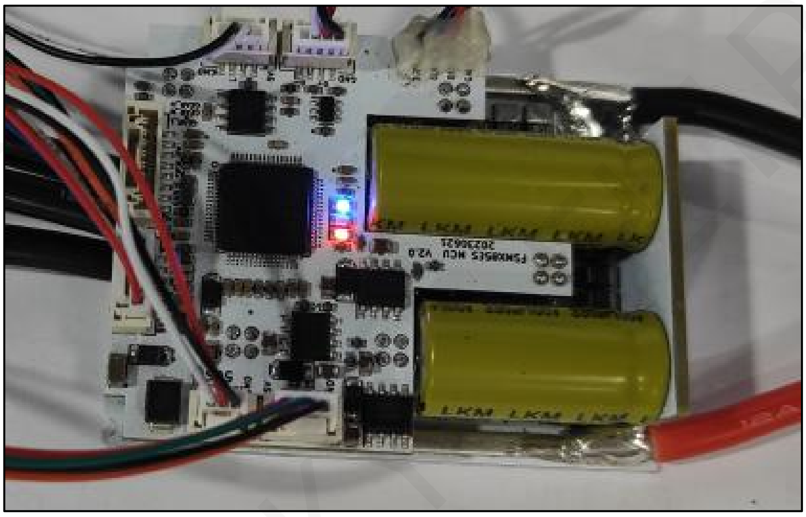
- Step 1: Click "Enter bootloader" and "Yes" to enter the Bootloader mode as below Screenshot 5-2.

Screenshot 5-2. Enter Bootloader Interface

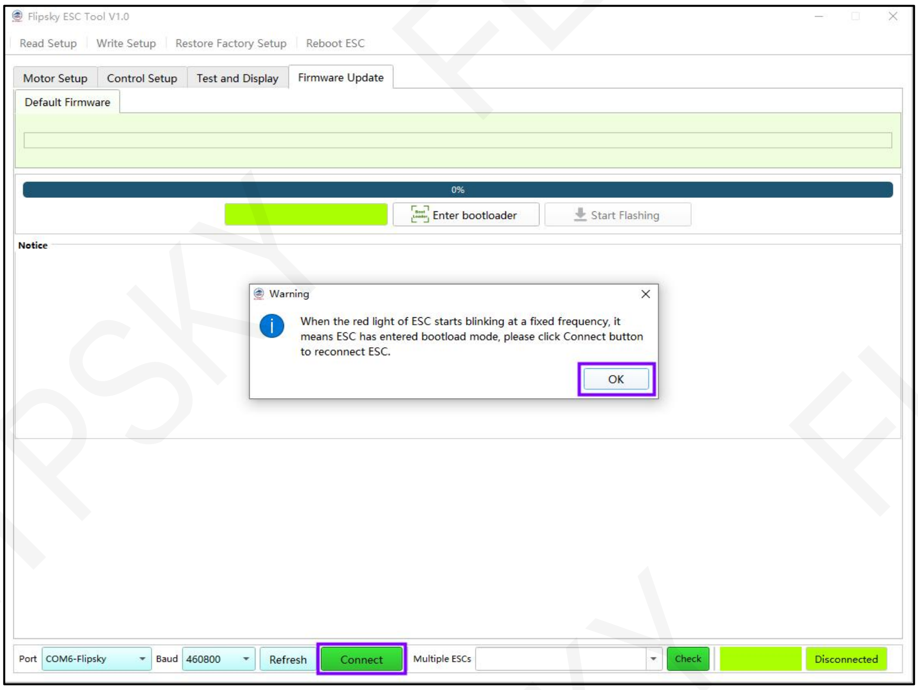


- Step 2: When the ESC red LED blinking at a fixed frequency, it means entering the Bootloader mode successfully (as shown in Screenshot 5-3). Click "OK" and "Connect" to reconnect the ESC as below Screenshot 5-4.

Screenshot 5-3. ESC entering the bootloader mode

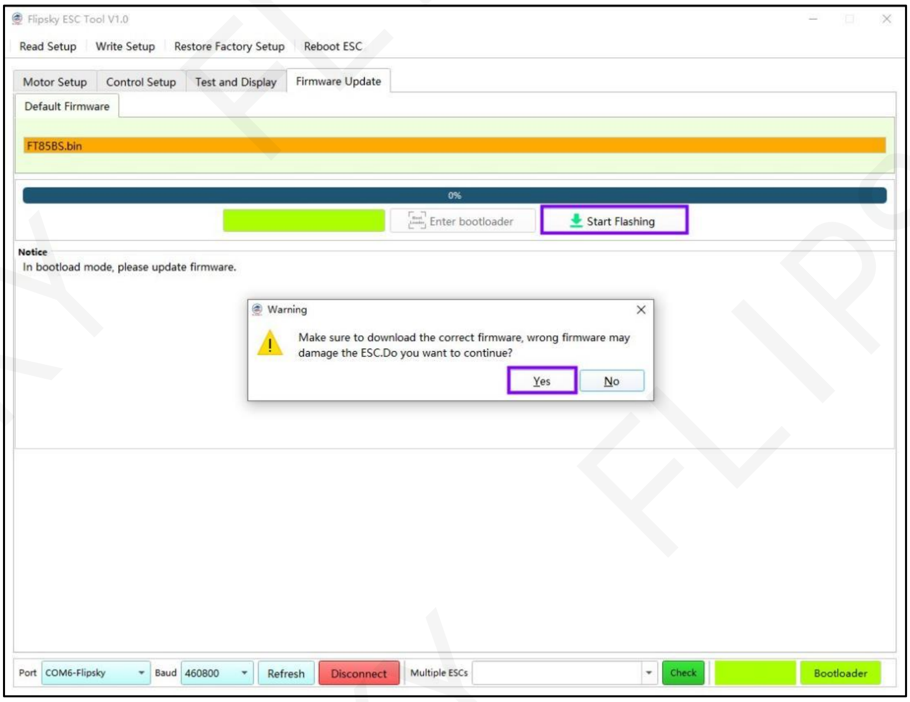


Screenshot 5-4. Reconnect the ESC

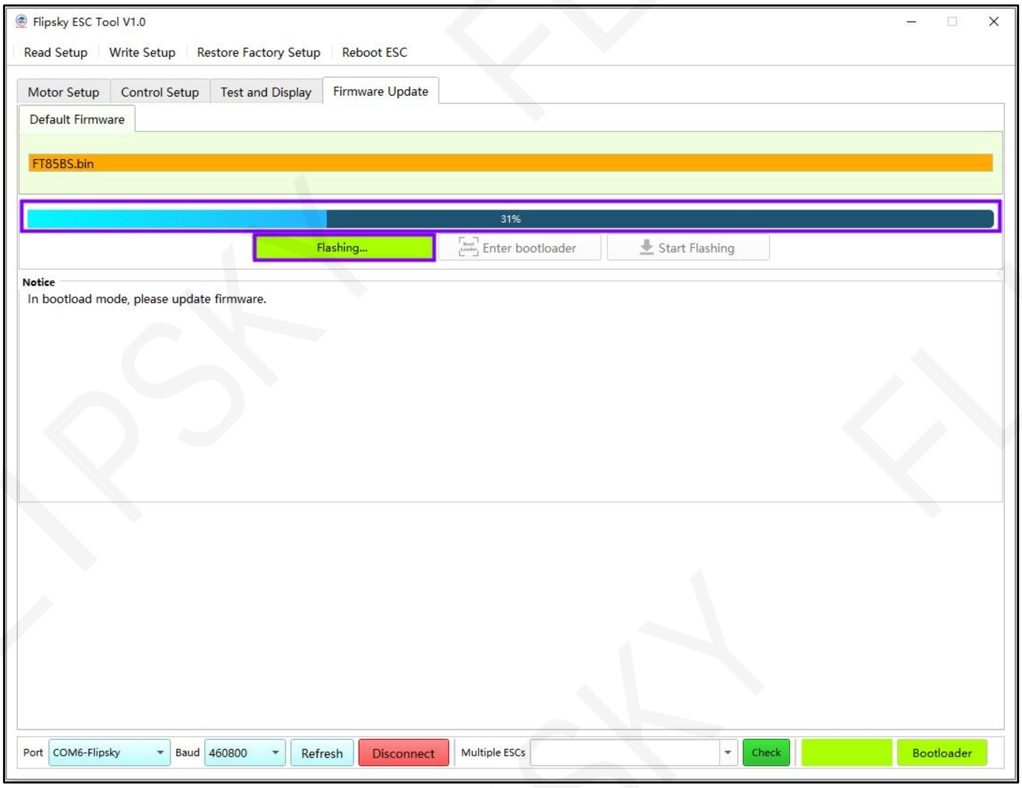


- Step 3: Click "Start Flashing" and "Yes" to update the firmware as below Screenshot 5-5 and Screenshot 5-6.

Screenshot 5-5. Start Flashing to update firmware

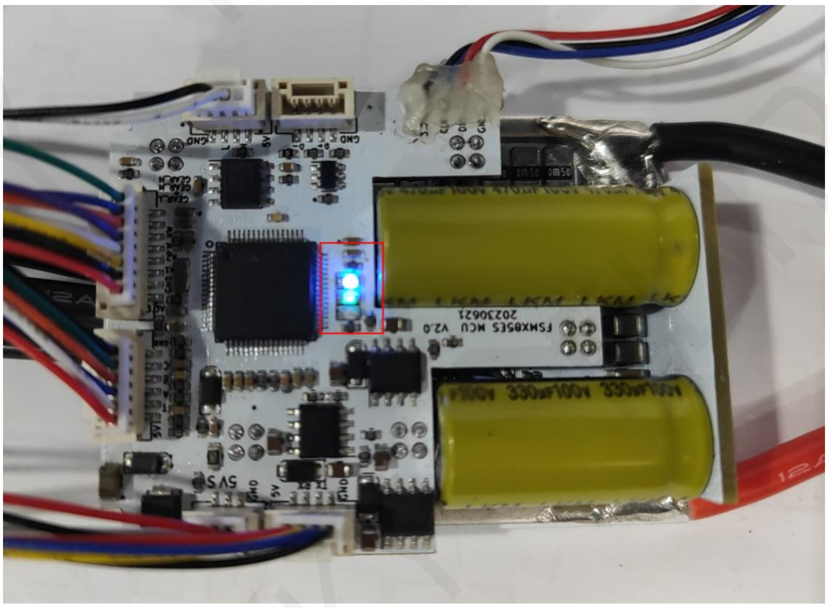


Screenshot 5-6. Firmware Update



- When the ESC's green and blue LEDs are solid on after the blinking of red and green LEDs, it means the finish of firmware update (as Screenshot 5-7). Click "Connect" to reconnect the ESC.

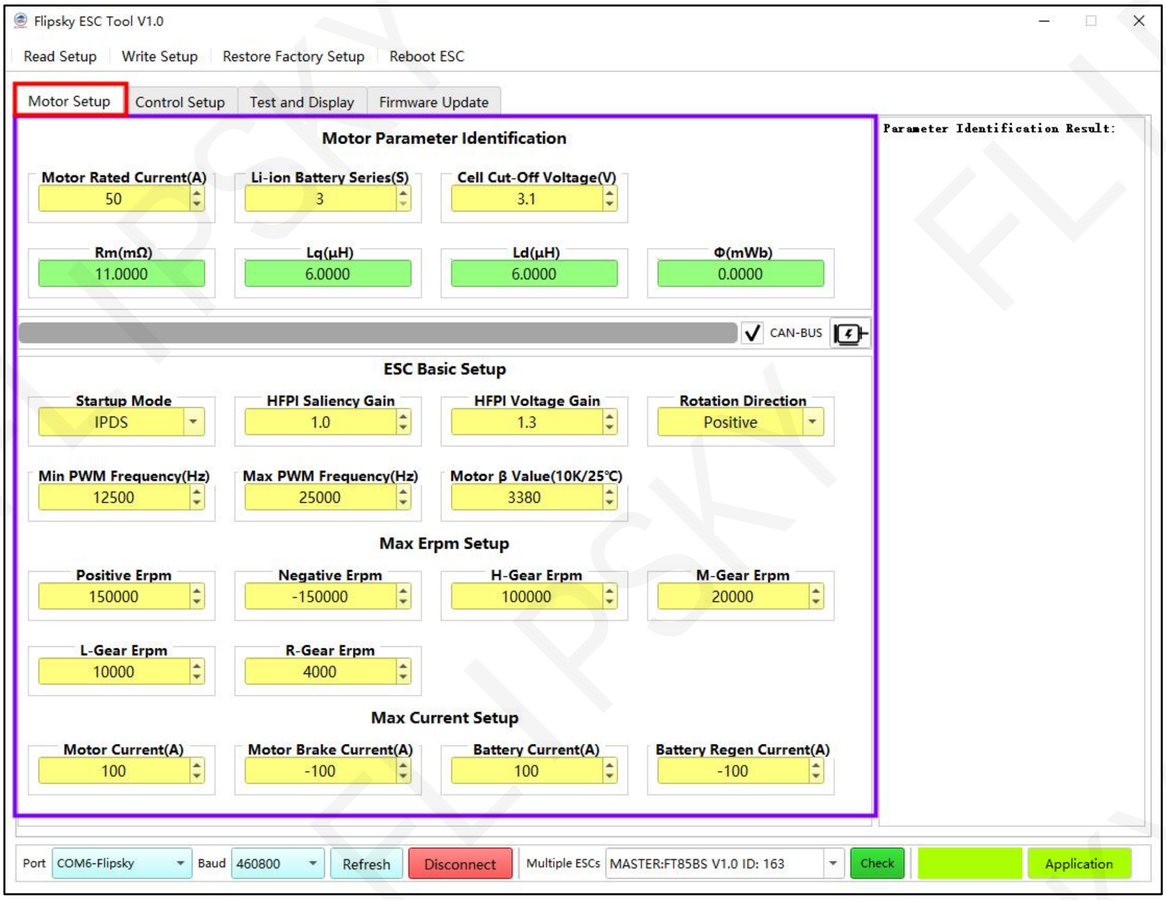
Screenshot 5-7. Solid blue and green led on



6 Motor Setup

Motor setup interface is in home interface as Screenshot 6-1.

Screenshot 6-1. Motor Setup Interface

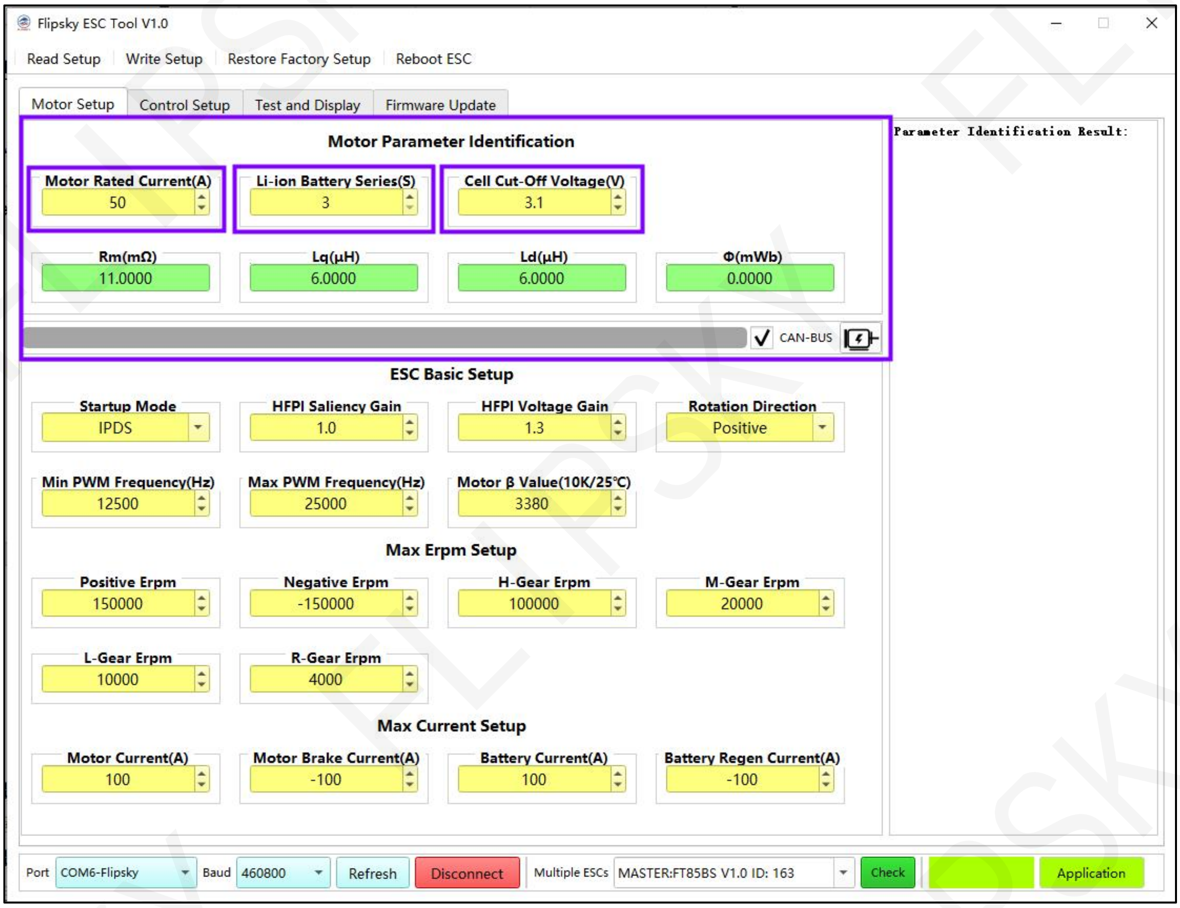


6.1 Parameters Identification Wizard

6.1.1 Preparations for Parameters Identification

Set "Motor Rated Current(A)", "Li-ion Battery Series(S)" and "Cell Cut-Off Voltage" before doing parameters identification as below Screenshot 6-2.

Screenshot 6-2. Three parameter settings before identification

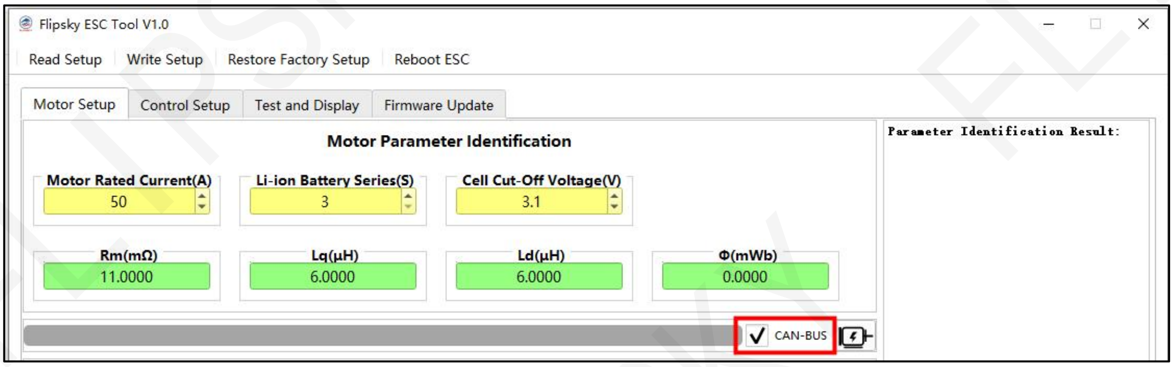


- Motor Rated Current(A): Write the motor rated current per motor specifications, not the max. or peak current (default: 50A; adjustable range: 5-500A)
- Li-ion Battery Series(S): Write the battery Lipos per battery specifications (1S=4.2V full charge; default: 3S; adjustable range: 3-30S)
- Cell Cut-Off Voltage(V): Battery voltage cutoff per cell setup to prevent the battery over discharge (default: 3.1V; adjustable range: 3.0 ~ 3.3V).

6.1.2 Parameters Identification Steps

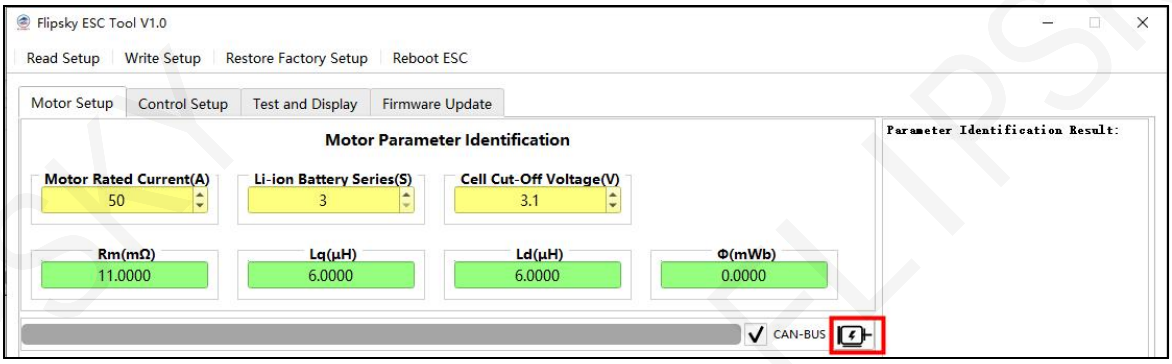
- Step 1: Set "Motor Rated Current(A)", "Li-ion Battery Series(S)", and "Cell Cut-Off Voltage(V)" accordingly (See Screenshot 6-2).
- Step 2: Remember to tick "CAN-BUS" when doing settings for dual or multiple ESCs and drivers as below Screenshot 6-3.

Screenshot 6-3 CAN-BUS

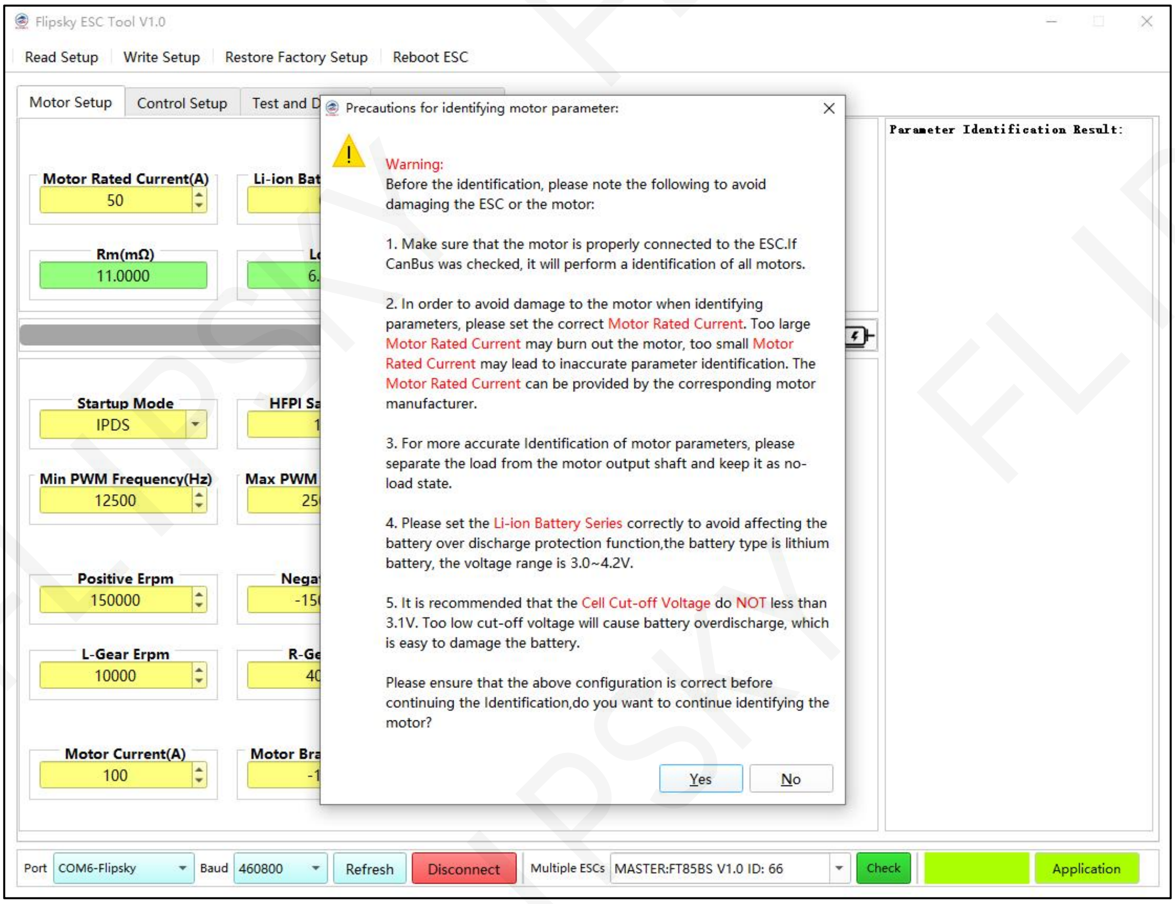


- Step 3: Click the Parameter Identification Button (as shown in Screenshot 6-4) and the notification window popup "Precautions for identifying motor parameter". Click “Yes” to proceed next step as below Screenshot 6-5.

Screenshot 6-4 Parameter Identification Button



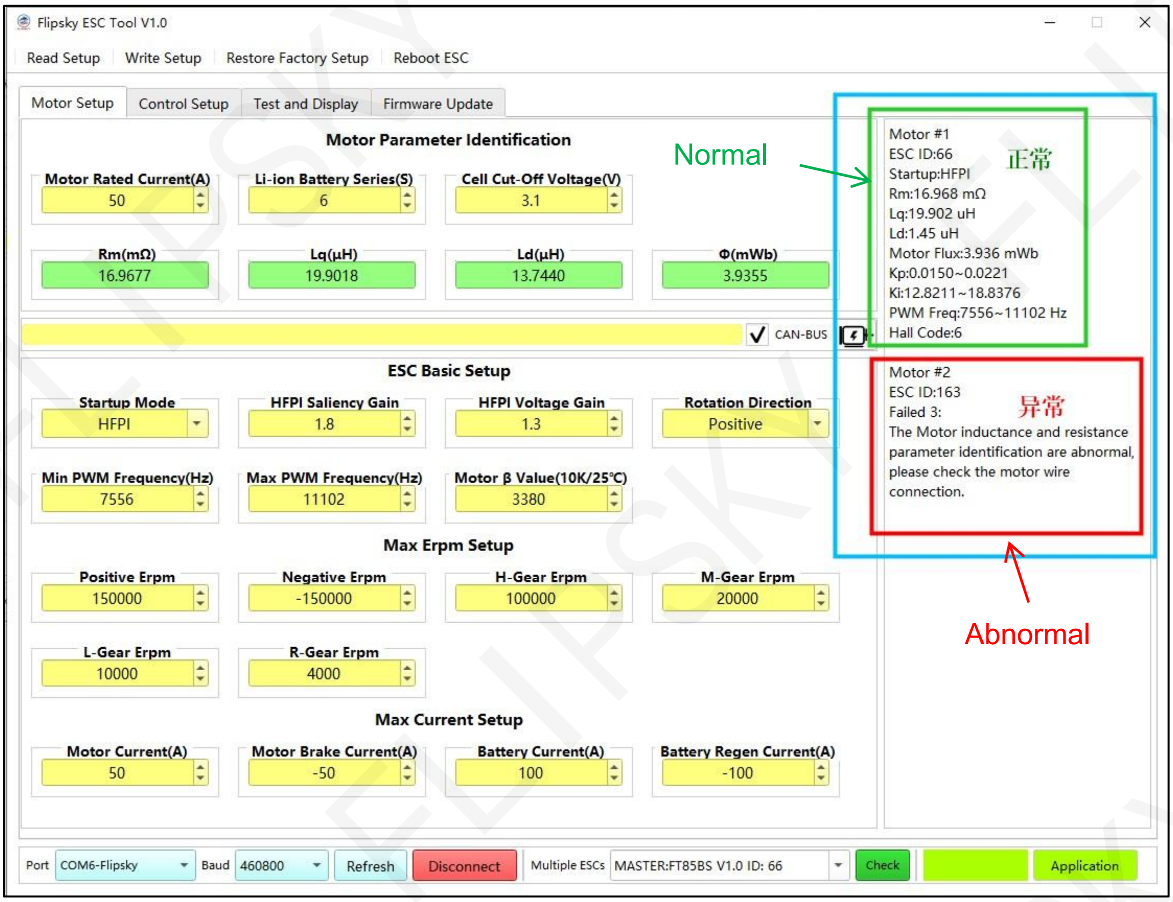
Screenshot 6-5 Precautions for identifying motor parameter



6.1.3 Motor Parameter Identification Results and Fault Codes

Please check the results on the right side of the home page (as shown in Screenshot 6-6) after the finish of motor Parameter Identification to see if all data are normal or not. If abnormal, please take corrective actions per fault codes and failed reasons notification.

Screenshot 6-6. Parameter identification results



Normal interpretation of motor parameter identification results:

- ESC ID: ESC ID Number
- Startup: Motor Starting Mode
    - ✧ IPDS: Initial Position Detection Startup (sensorless mode)
    - ✧ HFPI: High Frequency Pulse Injection (sensorless mode)
    - ✧ HALL: Hall Sensor Mode
- Rm: Motor Phase Resistance (mΩ)
- Lq: Motor Q-Axis Inductance (uH)
- Ld: Motor D-Axis Inductance (uH)
- Motor Flux(Φ): Motor Flux Linkage (mWb)

- Kp、Ki: Dynamic variation range of Proportional(Kp) and Integral(Ki)
- PWM Freq: Dynamic variation range of PWM Frequency
- Hall Code:
  - Hall Code: 1→HALL Sensor
  - Hall Code: 2→One Hall Phase Absence
  - Hall Code: 3→Two Hall Phase Absence
  - Hall Code: 4→Hall Three-Phase Short Circuit
  - Hall Code: 5→Overloading, Undetected
  - Hall Code: 6→No HALL Sensor Connection

Interpretation of motor parameters identification abnormal results:

- Failed 1: The battery Lipos number is set incorrectly, and does not match the actual operating voltage.Please set the correct number of battery Lipos.

  Notes: Please make sure the battery Lipos matching the using battery. Otherwise, it leads to detection failure, battery over-discharge protection, battery life span etc.
- Failed 2: The ESC is still faulty, please check the error code.

  Notes: Please check the fault code and take corrective actions per codes notification.
- Failed 3: The Motor inductance and resistance parameter identification are abnormal, please check the motor wire connection.

  Notes: Please check the ESC and motor wires connections.
- Failed 4: The motor hall sensor identify failed. please check the motor hall sensor.

  Notes: Please check the motor hall sensor wire connection and measure the motor hall sensor voltage to see if normal.
- Failed 5: The motor flux linkage identify failed. Maybe the motor load is too heavy.

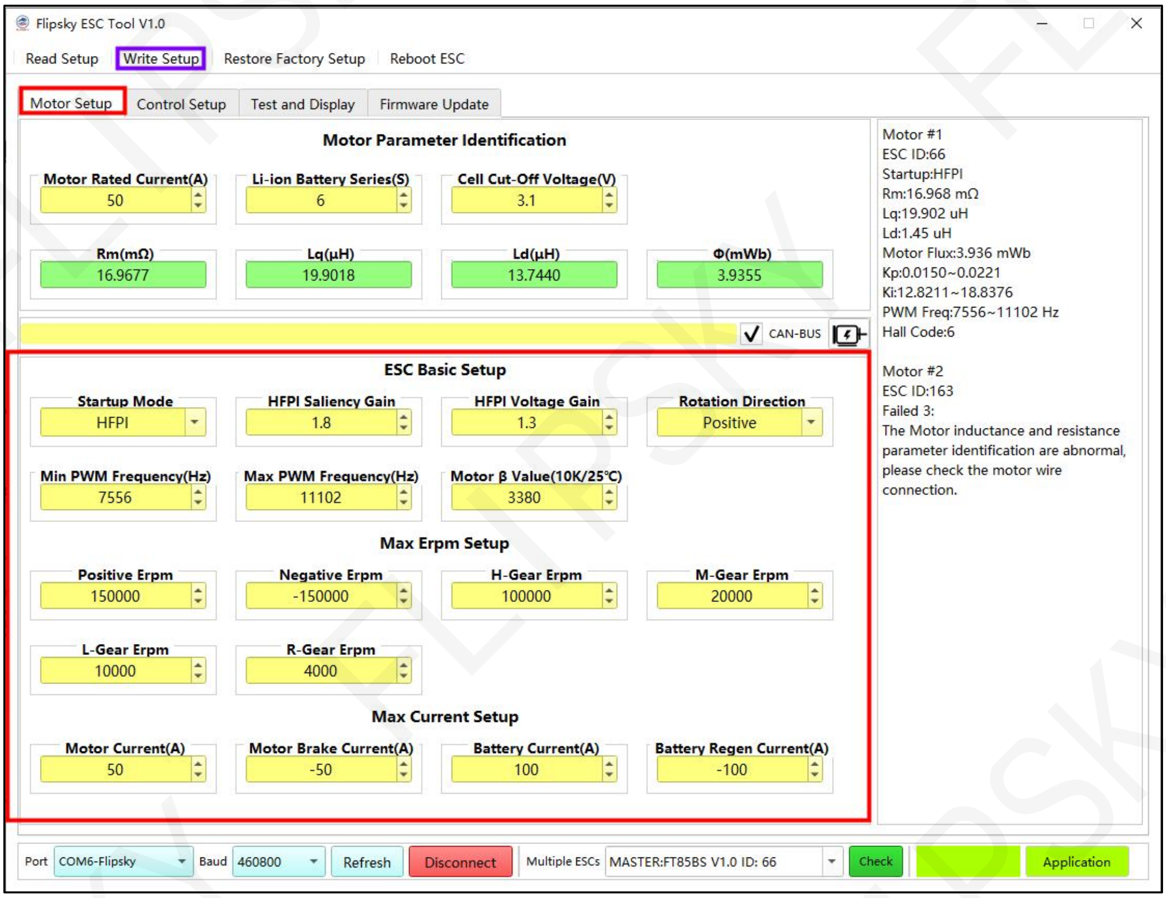
  Notes: Please check if overloading happens or the correct matching motor setup.

6.2 ESC Parameters Definitions and Configuration

Some motor parameters like Rm, Lq, Ld, Φ, Startup Mode、HFPI Saliency Gain、HFPI Voltage Gain、Max PWM Frequency 、 Min PWM Frequency etc. will be calculated automatically after the parameter identification finish which don't need to change manually. As for Max Erpm and Max Current, please write per application requirements as below Screenshot 6-7.

Notes: Remember to click the upper left button "Write Setup" to save settings.

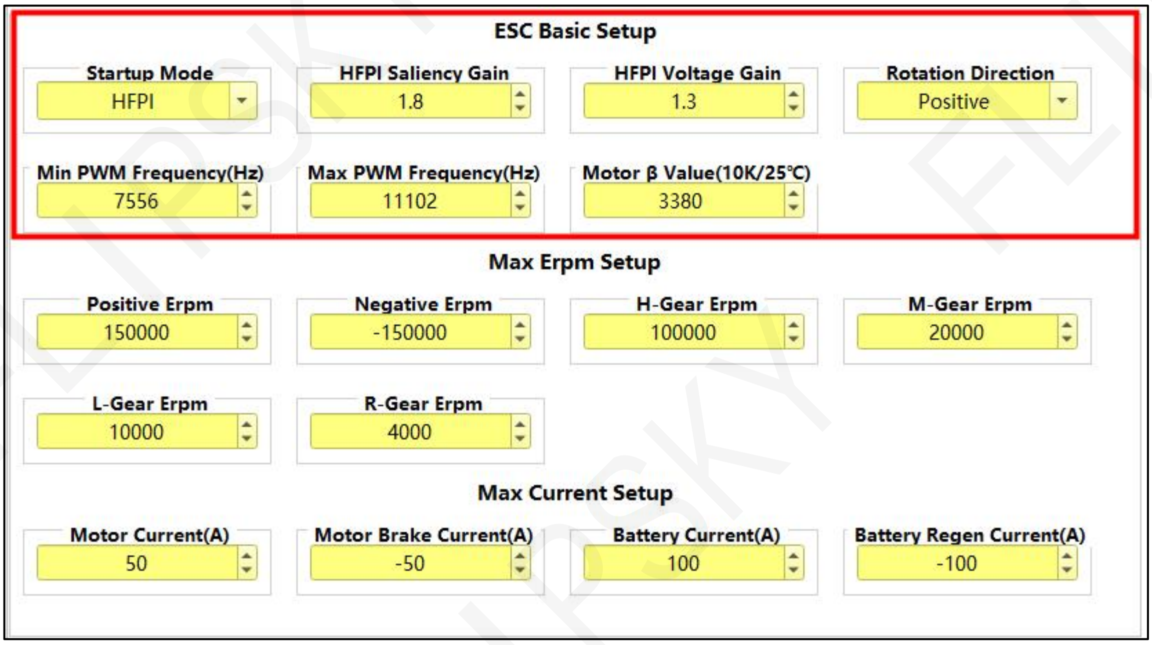
Screenshot 6-7. Adjustable parameters



6.2.1 ESC Basic Parameters

ESC basic parameters settings include "Startup Mode", "HFPI Saliency Gain", "HFPI Voltage Gain Volt", "Rotation Direction", "Min PWM Frequency", "Max PWM Frequency" and "NTC(25℃-10K) Beta Value" as below Screenshot 6-8:

Screenshot 6-8. ESC Basic Parameters



- Startup Mode

Three motor startup modes can be selected:

- ✧ IPDS: Initial Position Detection Startup (suitable for light loading sensorless mode such as underwater DPVs, multi-axis FPVs etc.)
- ✧ HFPI: High Frequency Pulse Injection (suitable for heavy loading sensorless mode such as E-Bike, E-Board, Go-cart etc.)
- ✧ HALL: Hall Sensored Mode (widely used, especially suitable for equipments with full load)

Note: HFPI mode is not suitable for all motors. If there are serious vibrations, stalls etc. when using this mode, it means that this mode is not suitable for the motor. If the motor integrates HALL interface, we strongly recommend using HALL for smooth running and better performance.

- **HFPI Saliency Gain**

Adjustable range: 0.5~4.0

Parameter description: This parameter setup is designed for the accuracy of angle measurement and reduce noise in HFPI mode, which can be matched automatically after the motor parameter identification finish and it’s not necessary to write manually.

- **HFPI Voltage Gain**

Adjustable range: 0.2~3.0

Parameter description: This parameter setup is designed for high-frequency pulse voltage injection needs of motor D-axis in HFPI mode, which is determined by the motor specification and can be matched automatically after the motor parameter identification finish, so it’s not necessary to write manually.

- **Rotation Direction**
    - ✧ Positive
    - ✧ Negative (Reverse)

- **Min PWM Frequency**

Adjustable range: 2000~45000

Parameter description: This parameter can be matched automatically after the motor parameter identification finish and no need to write manually.

- **Max PWM Frequency**

Adjustable range: 10000~45000

Parameter description: This parameter can be matched automatically after the motor parameter identification finish and no need to write manually.

- **Motor β Beta Value(25℃-10K)**

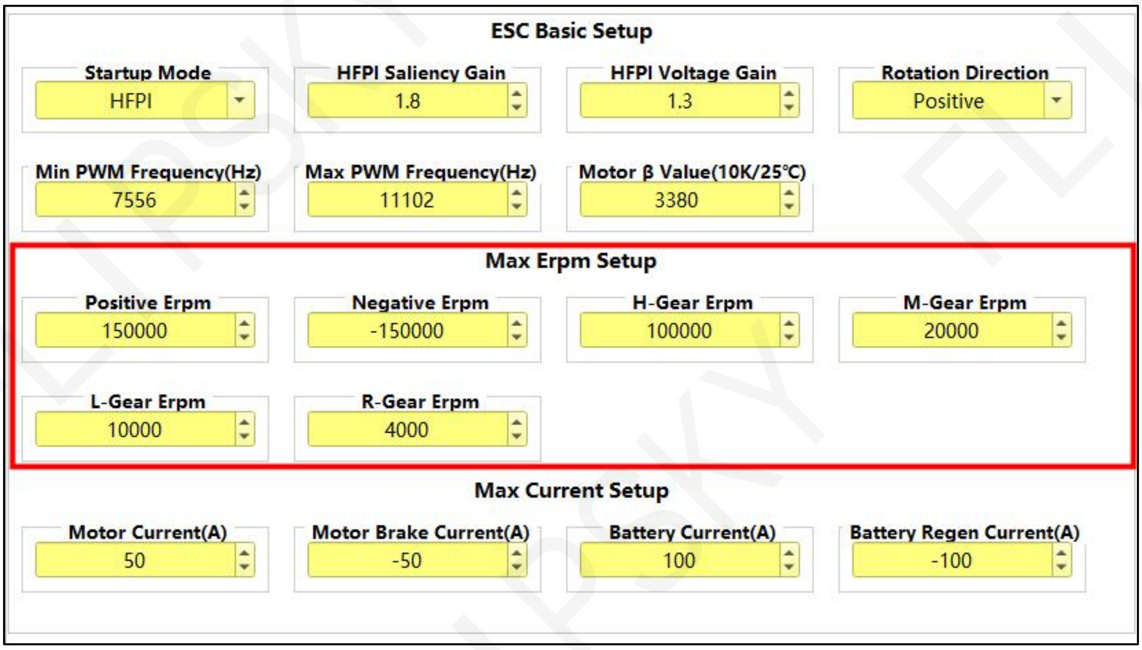
Adjustable range: 0~10000

Parameter description: It’s suitable for temperature sensor NTC 10K resistors at 25°C . All Flipsky motors with this temperature sensor, and the β value is 3380.

6.2.2 Max Erpm Setup

Max Erpm Setup includes "Positive Erpm", "Negative Erpm", "H-Gear Erpm", "M-Gear Erpm", "L-Gear Erpm", "R-Gear Max Erpm" as below Screenshot 6-9.

Screenshot 6-9. Max Erpm Setup

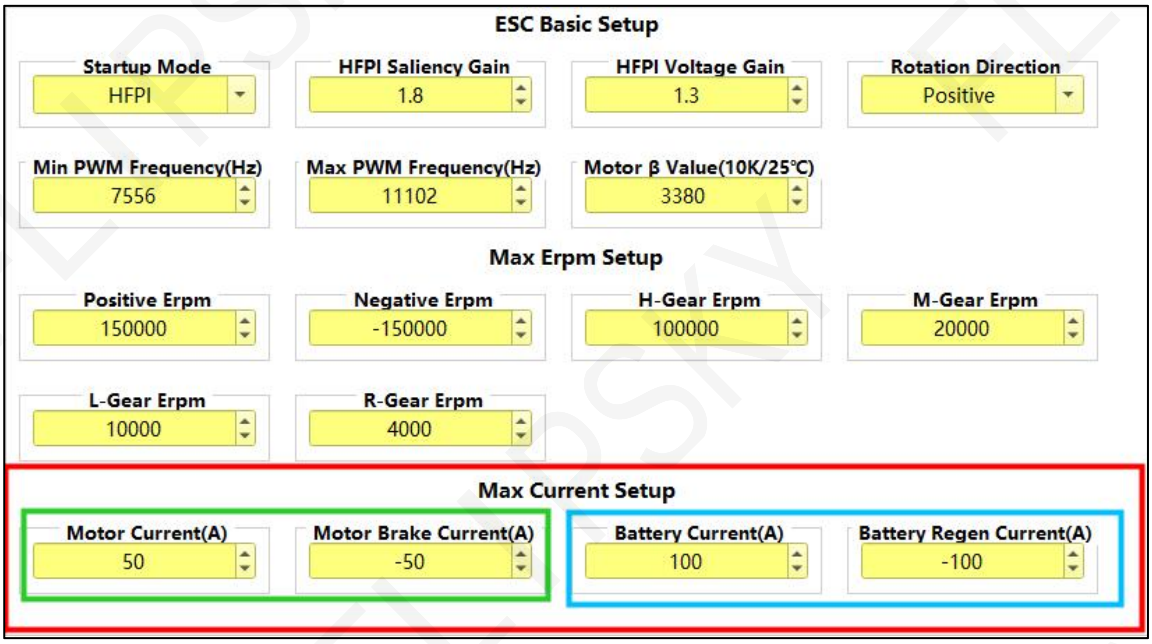


- Positive Max Erpm: Motor forward maximum Erpm
  Adjustable range: 500~250000
- Negative Max Erpm: Motor reverse maximum Erpm
  Adjustable range: -250000~-500
- H-Gear Max Erpm: The motor maximum Erpm at high speed gear
  Adjustable range: 500~250000
- M-Gear Max Erpm: The motor maximum Erpm at medium speed gear
  Adjustable range: 500~250000
- L-Gear Max Erpm: The motor maximum Erpm at low speed gear
  Adjustable range: 500~250000
- R-Gear Max Erpm: The motor maximum Erpm in reverse direction
  Adjustable range: 500~250000

6.2.3 Max Current Setup

The max current setup includes both the motor and the battery maximum current setup. The motor maximum current settings include "Motor Current" and "Motor Brake Current". Also, the battery maximum current settings include "Battery Current" and "Battery Regen Current" as below Screenshot 6-10.

Screenshot 6-10 Max Current Setup



- **Max Motor Current: The motor maximum continuous current (A)**
  Adjustable range: 0~500

- **Max Motor Brake Current: The motor maximum braking current (A)**
  Adjustable range: -500~0

Notes: If using the DC power supply, the Max Motor Brake Current must be set to **"0"**. Otherwise, the large back EMF will damage the ESC or even the power supply when the motor brakes in a high speed state.

- **Max Battery Current: Battery maximum continuous output current (A)**
  Adjustable range: 0~500

Notes: When setting Max Battery Current, it must be set according to the battery specification. Set a battery with maximum continuous output current 50A as an example:

single ESC, Max Battery Current: 50A

dual ESC, Max Battery Current: 50/2=25A

4WD ESC, Max Battery Current: 50/4=12.5A

The total output current of the battery must not exceed the battery maximum continuous output current specification. Otherwise, the BMS will cut off the power supply leading to motor's running out of control, which is very dangerous and may result in hurt for you. Moreover, please make sure that the Max Battery Current setting value does not exceed than that of the DC power supply to avoid ESC damage or unsafe happenings.

- **Max Battery Regen Current: battery max continuous regenerative current (A)**

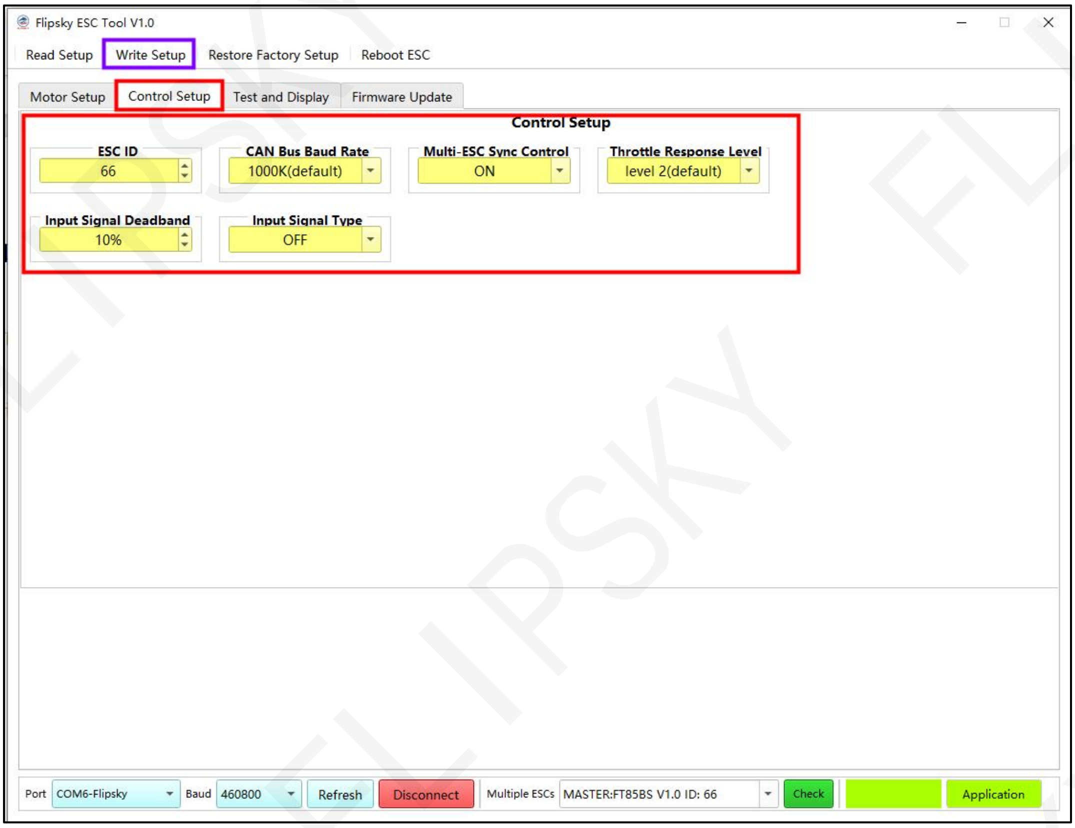
Adjustable range: -500~0

Note: If using the DC power supply, the Max Battery Regen Current must be set to **"0"**. Otherwise, the large back EMF will damage the ESC or even the power supply when the motor brakes in a high speed mode.

7 Control Setup

The input control setup includes "ESC ID", "CAN Bus Baud Rate", "Multi-ESC Sync Control", "Throttle Response Level", "Input Signal Deadband", "Input Signal Type" as below Screenshot 7-1.

Screenshot 7-1.Control Setup Interface

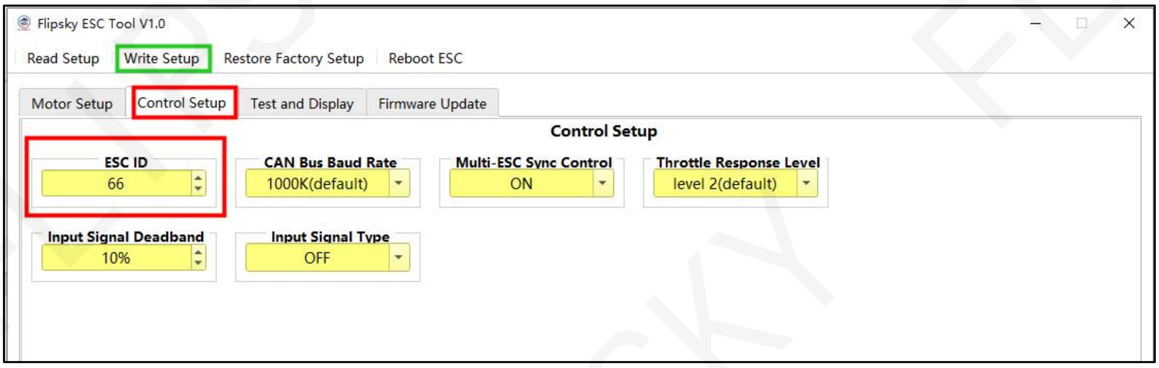


7.1 ESC ID Setup

ESC ID: ESC ID Number (See screenshot 7-2) adjustable range: 1~254 for multiple ESCs via CAN. Please make sure the uniqueness of the ESC ID since the same ESC IDs result in parameter identification failures or abnormal control. ESC ID 0 (special command number) and ESC ID 255 (special broadcast command) are not available for setup.

ESC ID number setup as Screenshot 4.2.1.

Screenshot 7-2.ESC ID

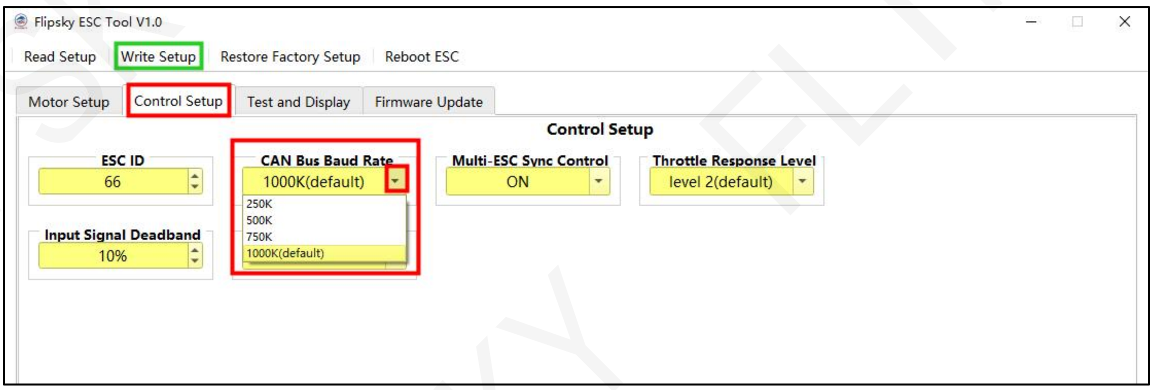


7.2 CAN-Bus Baud Rate Setup

CAN-Bus Baud Rate (see as Screenshot 7-3): Please ensure the consistency of CAN baud rate for the same CAN devices.

CAN-Bus baud rate setup as 4.2.2.

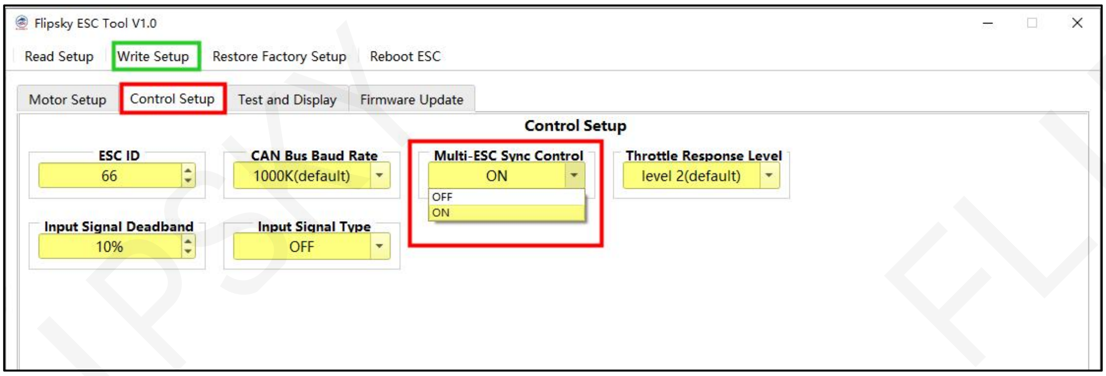
Screenshot 7-3 CAN-Bus Baud Rate



7.3 Multiple ESCs Sync Control Setup

Multi-ESC Sync Control: synchronous control for multiple ESCs via CAN as screenshot 7-4.

Screenshot 7-4.CAN Multiple ESCs



Setting steps as below:

- Click “Control Setup” to enter the setup interface.
- Click "Multi-ESC Sync Control" to select whether synchronous control of the CAN ESCs or not. (ON means Synchronous Control, OFF means not)
- Click the upper left menu "Write Setup" to save the setup

Parameter Description:

When the ESC is the master board and the "Multi-ESC Sync Control" is set to "ON", the input signal will synchronously control the master ESC and all CAN slave ESCs. If the "Multi-ESC Sync Control" is set to "OFF" in this situation, then the input signal controls the master ESC only.

When the ESC is the slave board and the "Multi-ESC Sync Control" is set to "ON", this slave ESC will under the control of master CAN-Bus. If the "Multi-ESC Sync Control " is set to "OFF" in this case, the slave ESC will shield the control commands from the master CAN-Bus.

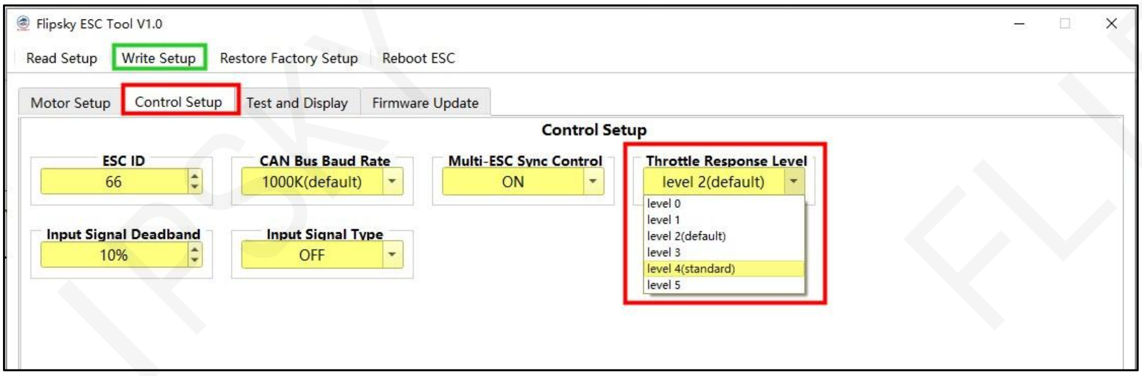
Therefore, if you want multiple ESCs to be controlled synchronously, set all ESC’s Multi-ESC Sync Control “ON”. However, if you want to control each ESC seperately in UART, PPM or ADC control type, set all ESC’s Multi-ESC Sync Control “OFF”.

If use your own control module as the master board and want to control the ESCs via CAN-Bus, set all ESC’s Multi-ESC Sync Control "ON", and the Input Signal Type "OFF".

7.4 Throttle Response Level Setup

Throttle Response Level: Level 0-Level 5 (6 level choices as screenshot 7-5).

Screenshot 7-5 Throttle Response Level Setup



Setting steps as below:

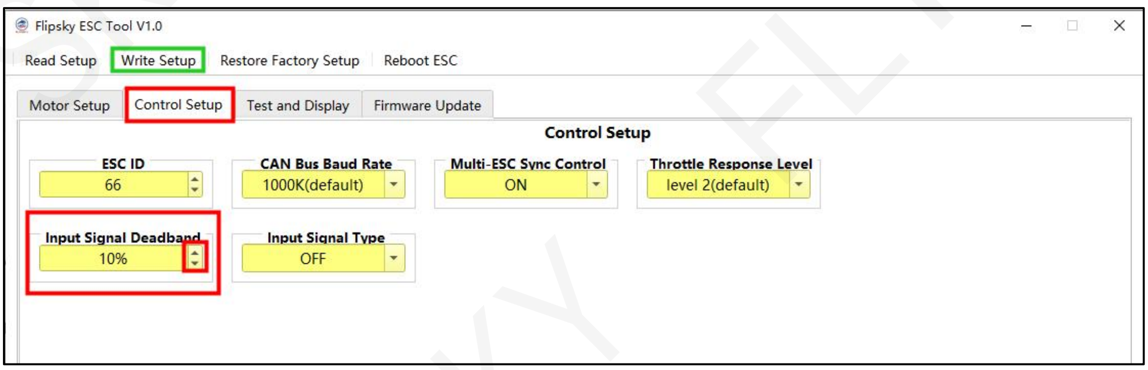
- Click “Control Setup” to enter the setup interface.
- Click "Throttle Response Level" to select the appropriate throttle response level.
- Click the upper left menu "Write Setup" to save the setup.

Parameter description: The higher of the level, the more sensitive of the throttle and brake response. The default level is Level 2, and Level 4 is the standard level which has no lead or lag on all input signals and is suitable for customers customized requirement.

7.5 Input Deadband Setup

Input Signal Deadband：Adjustable range: 1%-25% as below screenshot 7-6:

Screenshot 7-6. Input Signal Deadband Setup



Setting steps as below:

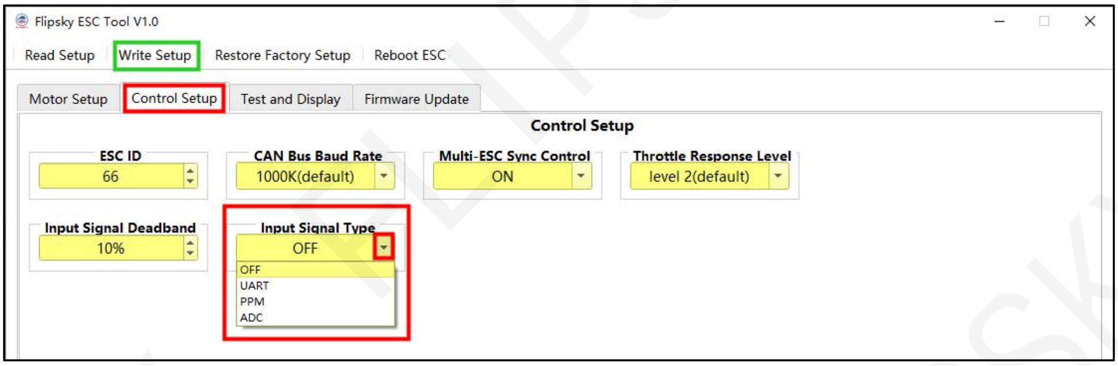
- Click “Control Setup” to enter the setup interface.
- Click "Input Signal Deadband" to adjust the appropriate percentage;
- Click the upper left menu "Write Setup" to save the setup.

Parameter description: The ESC starts the current output only when the throttle input signal percentage is over than that of the input deadband setup . It’s recommended to use the default value 10%. Regarding the input control type PPM and UART, the deadband percentage can be small or even 0% as the remote controller has set already, for example, the Flipsky remote controller VX4 with 5% deadband already . It’s also fine to keep the default setting 10% as the ADC control type recommended.

7.6 Input Control Type Setup

The input signal control type: OFF, UART, PPM and ADC (4 choices) and the default setting is OFF as below screenshot 7-7.

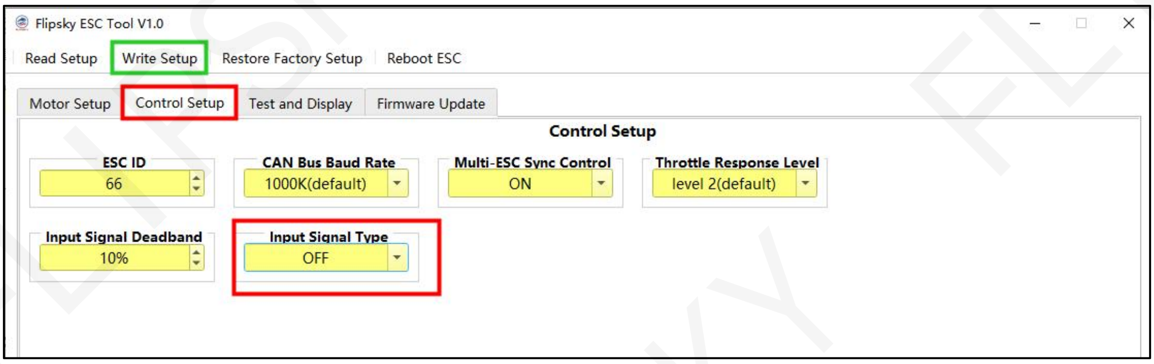
Screenshot 7-7. Input Signal Control Type



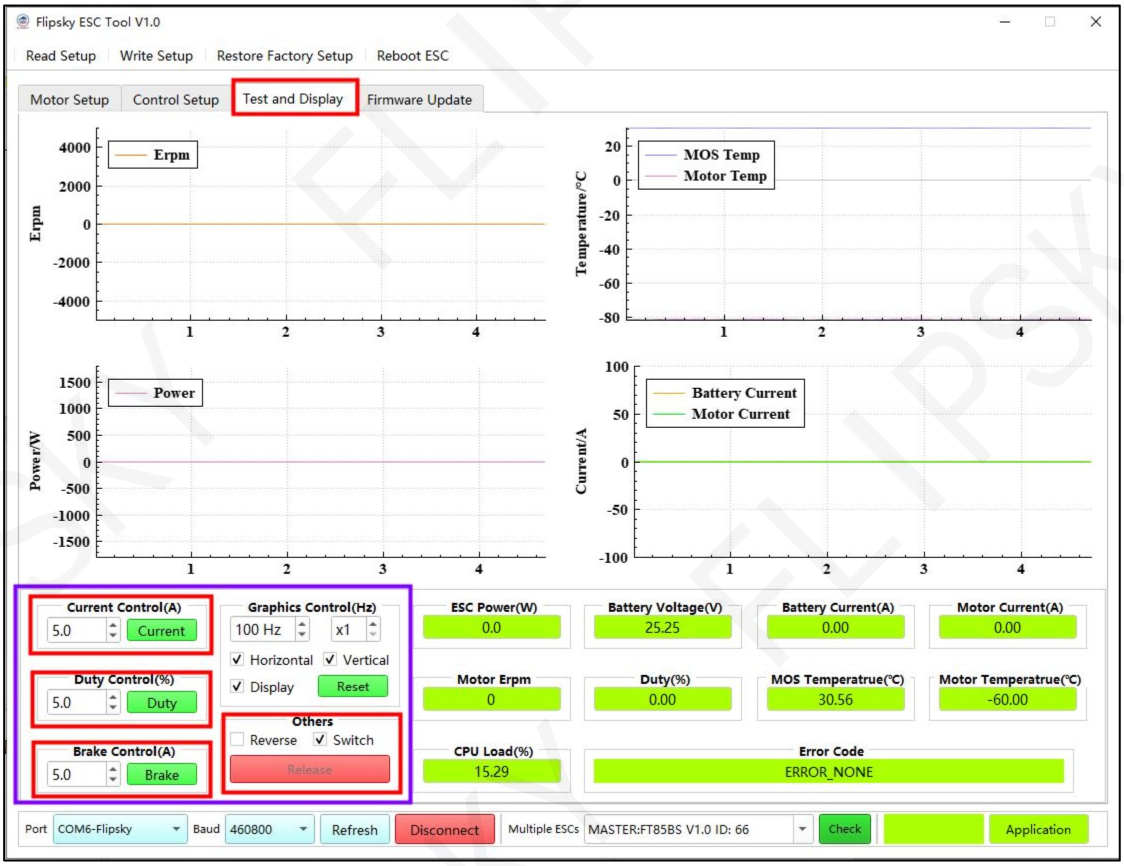
7.6.1 OFF

If the "Input Signal Type" is set to "OFF" (Screenshot 7-8), it means that all input signals control will be shielded. The ESC can be controlled through the Test and Display interface button control only like "Current Control", "Duty Control" and "Brake Control" (Screenshot 7-9). Please refer to 8.1.1 for button control setup.

Screenshot 7-8 Input Signal Type OFF



Screenshot 7-9.Test and Display Interface

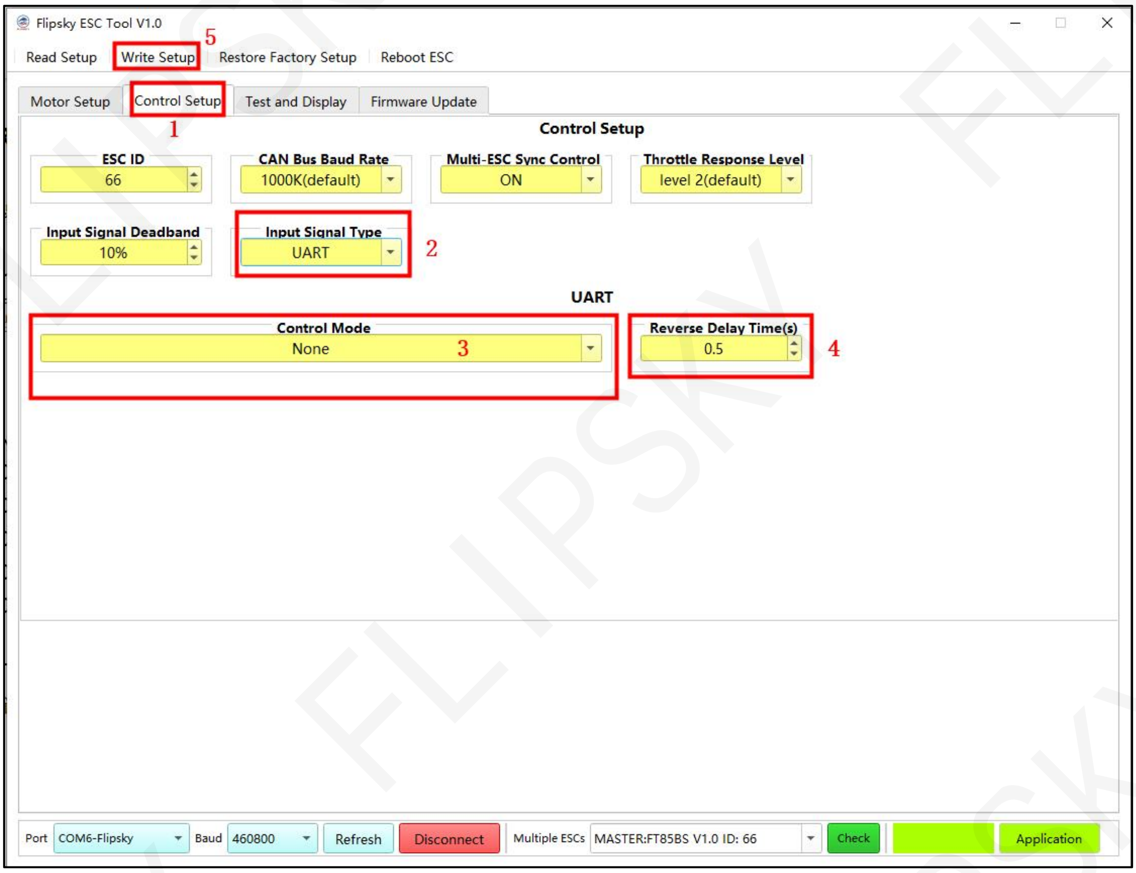


7.6.2 UART

7.6.2.1 UART Description and Setup

The UART mode supports Flipsky remote controller VX4's latest firmware VX4_V1_2 only. The max EPRM can be set under different gear speed levels for this mode.

Screenshot 7-10 UART Setup Interface



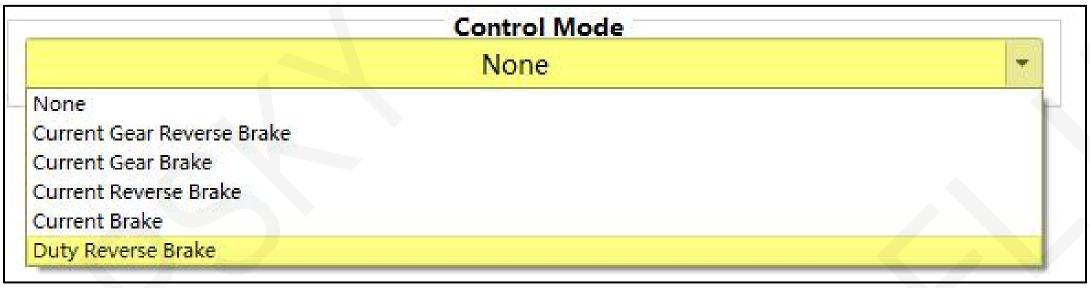
Setting steps as below:

- Step 1: Click "Control Setup" to enter the setup interface.
- Step 2: Switch the "Input Signal Type" to "UART"
- Step 3: Select the appropriate UART Control Mode per your needs (see details in 7.6.2.2)
- Step 4: Set the Reverse Delay Time (see details in 7.6.2.3)
- Step 5: Click the upper left menu "Write Setup" to save the setup.

7.6.2.2 UART Control Mode Setup

Select the control mode per your application needs as below screenshot 7-11.

Screenshot 7-11.UART Control Mode

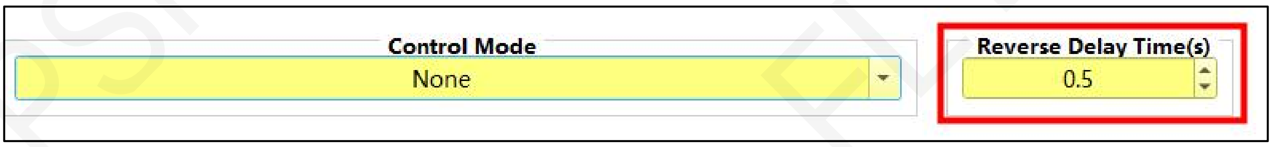


- None
- Current Gear Reverse Brake: Current control mode: gear speed level limit, reverse with brake, reverse
- Current Gear Brake: Current control mode: gear speed level limit, reverse with brake
- Current Reverse Brake: Current control mode: reverse with brake, reverse
- Current Brake: Current control mode: reverse with brake
- Duty Gear Brake Reverse: Dutycycle control mode: gear speed level limit, reverse with brake, reverse

7.6.2.3 UART Reverse Delay Time Setup

If the control mode choice of UART with reverse, click "UART Reverse Delay Time" for reverse response time setup. Adjustable range: 0.10s ~ 5.00s (default 0.5s) as below screenshot 7-12.

Screenshot 7-12 UART Reverse Delay Time Setup



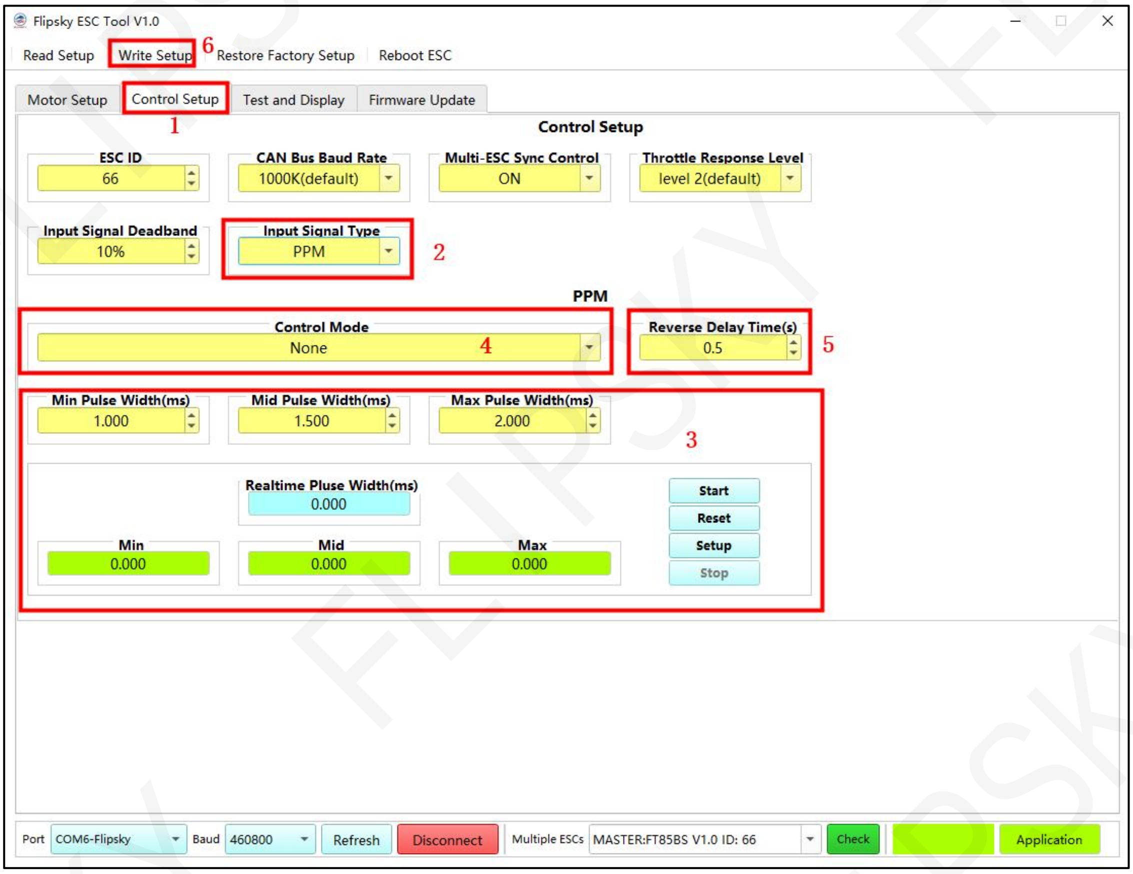
Parameter description: When the motor stops running, max pull or push the throttle gear for reverse as the Reverse Delay Time setup period, then the motor will run in reverse direction as screenshot 7.12.

7.6.3 PPM

7.6.3.1 PPM Description and Setup

The PPM signal pulse width usually ranges from 1ms to 2ms, so theoretically the PPM control frequency supports up to 500Hz. Considering the MCU clock accuracy, it is recommended to use the PPM control frequency less than 400Hz.

Screenshot 7-13 PPM Input Setup



Setting steps as below:

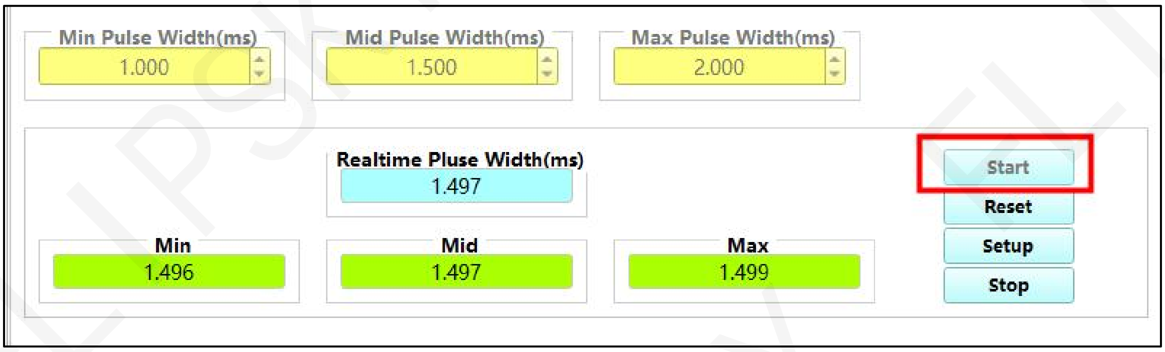
- Step 1: Click “Control Setup” to enter the setup interface.
- Step 2: Switch the "Input Signal Type" to "PPM"
- Step 3: Do PPM throttle calibration for the first time as details in 7.6.3.2
- Step 4: Select the appropriate PPM Control Mode per your needs (see details in 7.6.3.3)
- Step 5: Set the Reverse Delay Time (see details in 7.6.3.4)
- Step 6: Click the upper left menu "Write Setup" to save the setup.

7.6.3.2 PPM Input Throttle Calibration

Please do PPM input setup for the first time application. Setting Flipsky Remote VX1 as an example:

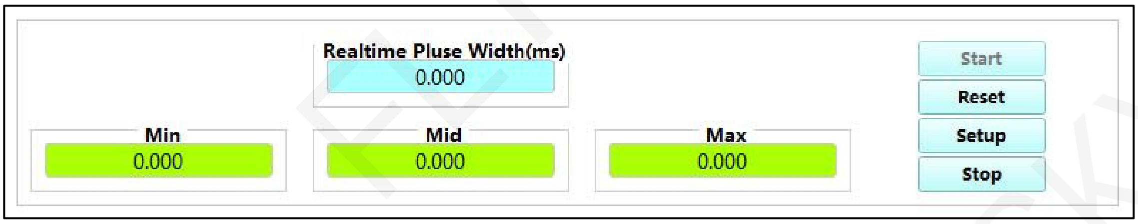
- Click "Start" to begin the calibration as below screenshot 7-14.

Screenshot 7-14 Start PPM Input Setup



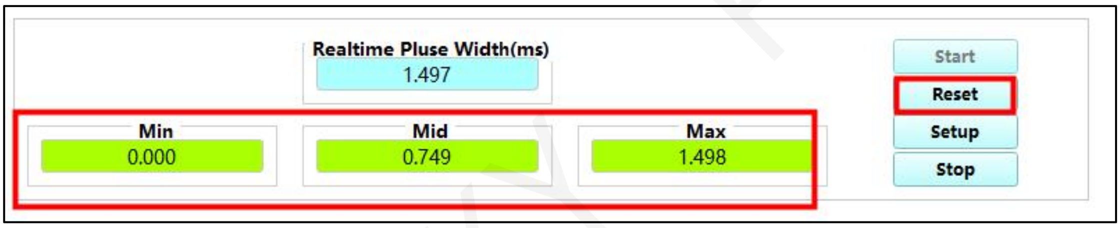
Notes: No PPM signal can be detected if the remote receiver hasn’t been plugged with the ESC PPM wire when you click "Start" as below screenshot 7-15.

Screenshot 7-15. No PPM Signal Data



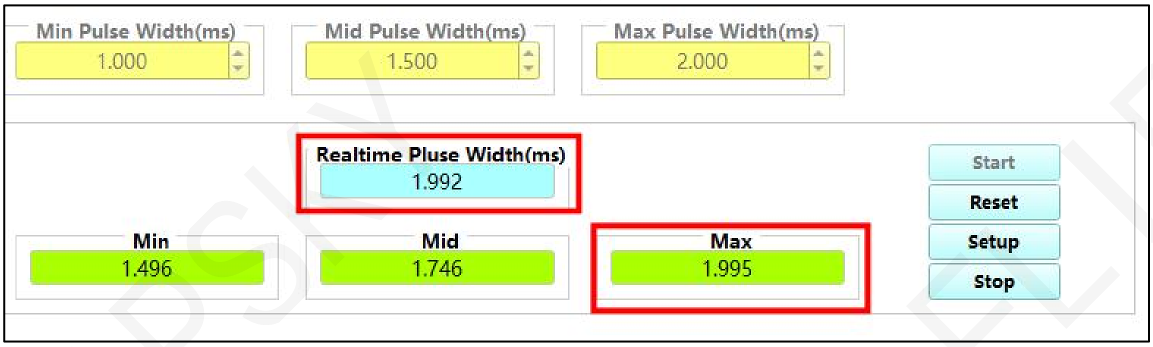
- Connect the remote receiver with the ESC PPM wire in this case. If the PPM signal data seems abnormal (as shown in Screenshot 7-16), please click "Reset". (Refer to Screenshot 7-16).

Screenshote 7-16.PPM Signal Data Abnormal

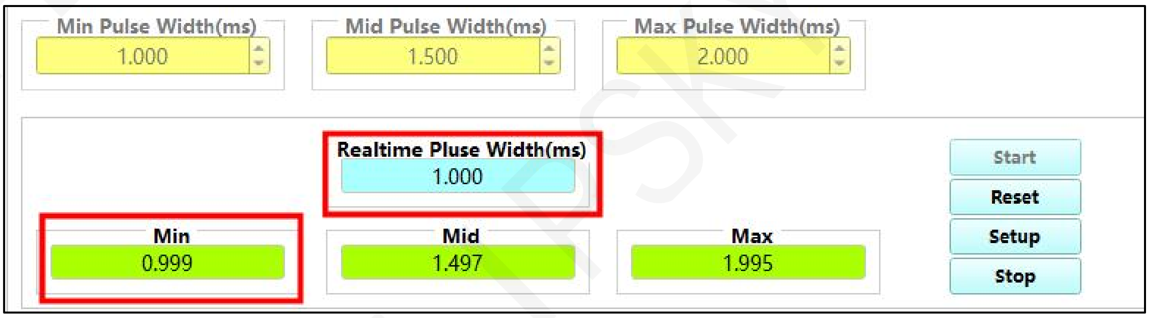


■ Max pull and push the remote throttle gear to refresh the minimum, middle and maximum PPM signal values as below screenshot 7-17 and screenshot 7-18.

Screenshot 7-17. PPM signal max.



Screenshot 7-18. PPM signal min.

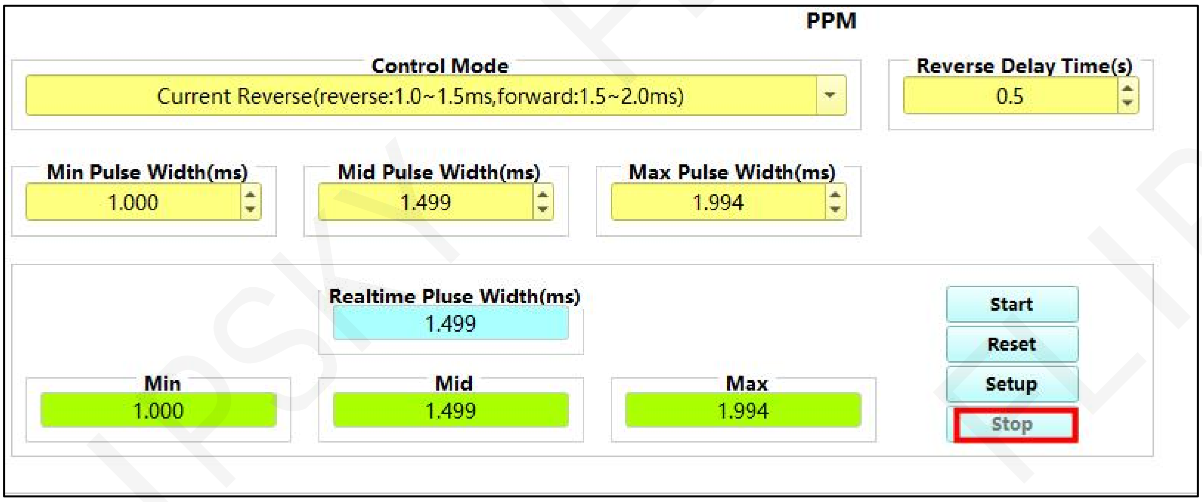


■ Click "Setup" to write the "Min", "Mid" and "Max" data to Pulse Width accordingly as screenshot 7-19.

Screenshot 7-19. Set PPM Pulse Width Values

- Click "Stop" to complete the PPM input setup as screenshot 7-20.

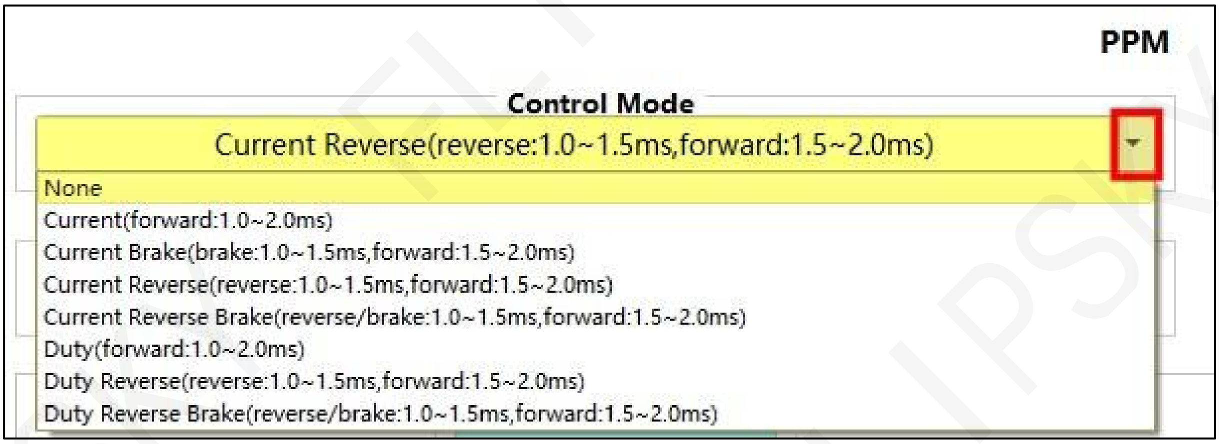
Screenshot 7-20.PPM Input Setup Finish



7.6.3.3 PPM Control Mode Setup

Select the appropriate PPM Control Mode per your needs as screenshot 7-21:

Screenshot 7-21.PPM Control Mode Setup



- None
- Current (forward:1.0~2.0ms)

  Current control mode, unidirectional control, only forward:1.0~2.0ms.
- Current Brake (brake:1.0~1.5ms,forward:1.5~2.0ms)

  Current control mode, bidirectional control, brake:1.0~1.5ms, forward:1.0~2.0ms.
- Current Reverse (reverse1.0~1.5ms, forward:1.5~2.0ms)

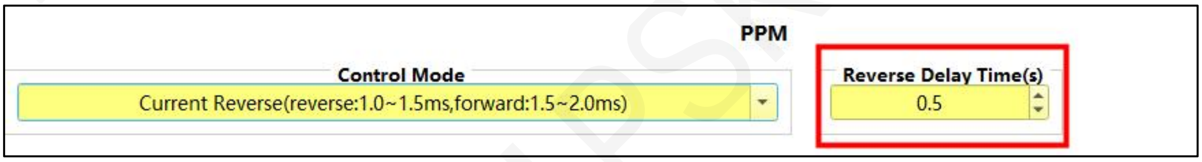
  Current control mode, bidirectional control, no brake, backward:1.0~1.5ms, forward:1.0~2.0ms.

- Current Reverse Brake(reverse/brake:1.0~1.5ms, forward:1.5~2.0ms)
  Current control mode, bidirectional control, brake/reverse:1.0~1.5ms, forward:1.0~2.0ms.
- Duty(forward:1.0~2.0ms)
  Duty control mode, unidirectional control, only forward:1.0~2.0ms.
- Duty Reverse(reverse:1.0~1.5ms, forward:1.5~2.0ms)
  Duty cycle control mode, bidirectional control, backward:1.0~1.5ms, forward:1.0~2.0ms.
- Duty Reverse Brake(reverse/brake:1.0~1.5ms, forward:1.5~2.0ms)
  Duty cycle control mode, bidirectional control, brake/reverse:1.0~1.5ms, forward:1.0~2.0ms.

7.6.3.4 PPM Reverse Delay Time Setup

If the control mode choice of PPM with reverse, click PPM “Reverse Delay Time" for reverse response time setup. Adjustable range: 0.10s ~ 5.00s (default 0.5s) as below screenshot 7-22:

Screenshot 7-22 PPM Reverse Delay Time Setup



Parameter description: When the motor stops running, max pull or push the throttle gear for reverse as the Reverse Delay Time setup period, then the motor will run in reverse direction.

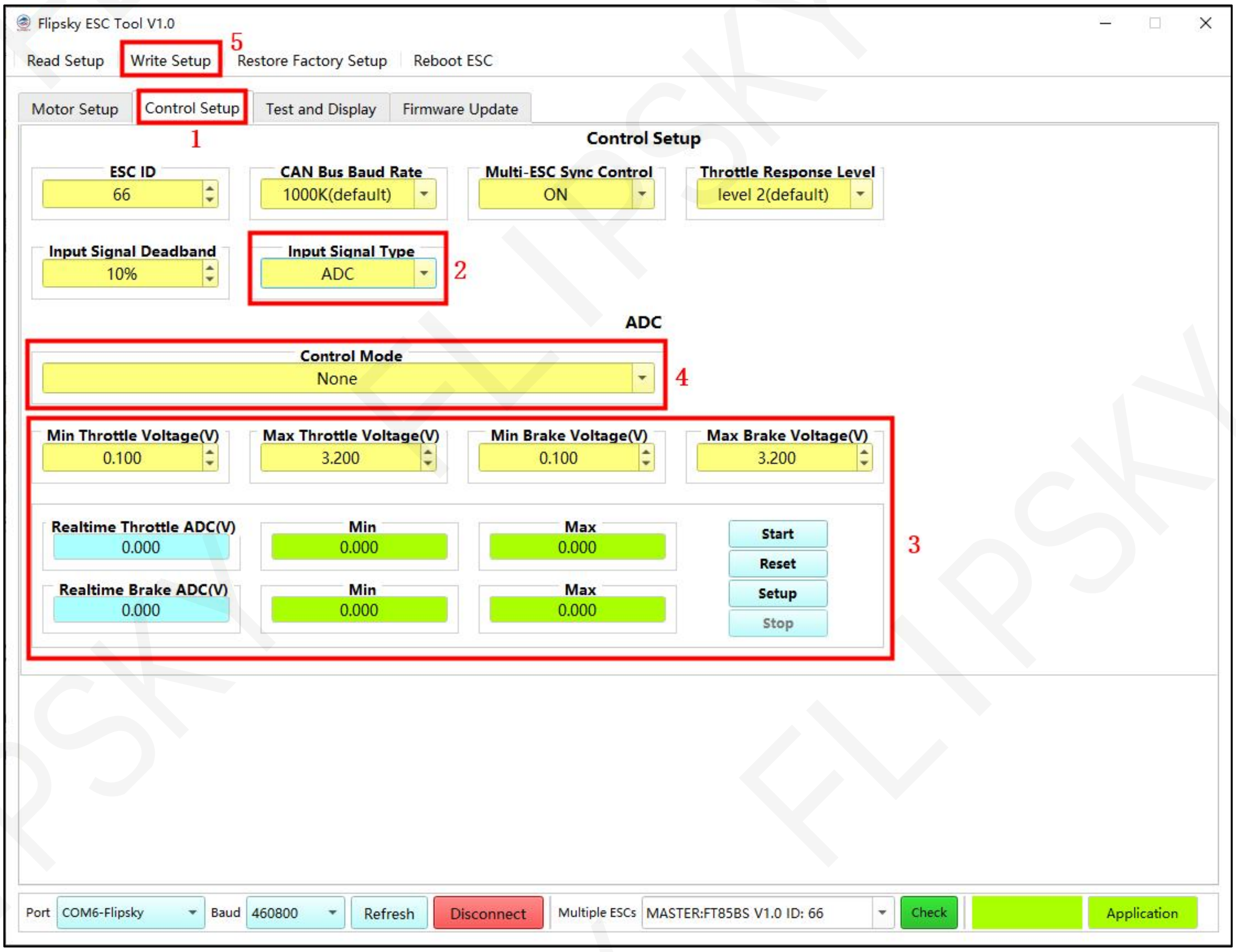
7.6.4 ADC

7.6.4.1 ADC Description and Setup

ADC control type voltage range: 0V~3.3V. Please make sure the ADC input voltage not over than 3.3V to avoid the ESC 3.3V chip damage. Setting steps as below screenshot 7-23.

- Step 1: Click “Control Setup” to enter the setup interface.
- Step 2: Switch the "Input Signal Type" to "ADC"
- Step 3: Do ADC throttle calibration for the first time as details in 7.6.4.2
- Step 4: Select the appropriate ADC Control Mode per your needs (see details in 7.6.4.3)
- Step 5: Click the upper left menu "Write Setup" to save the setup.

Screenshot 7-23.ADC Control Type Setup

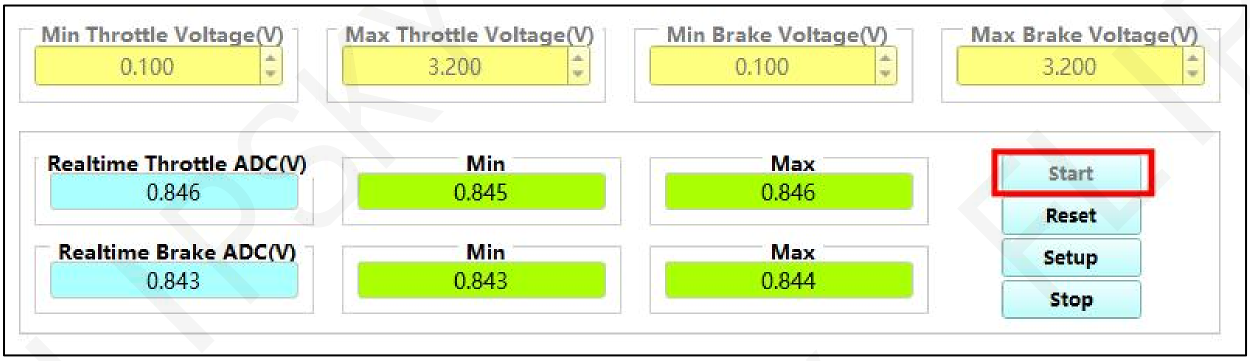


7.6.4.2 ADC Input Throttle Calibration

Do ADC throttle calibration for the first time as follows:

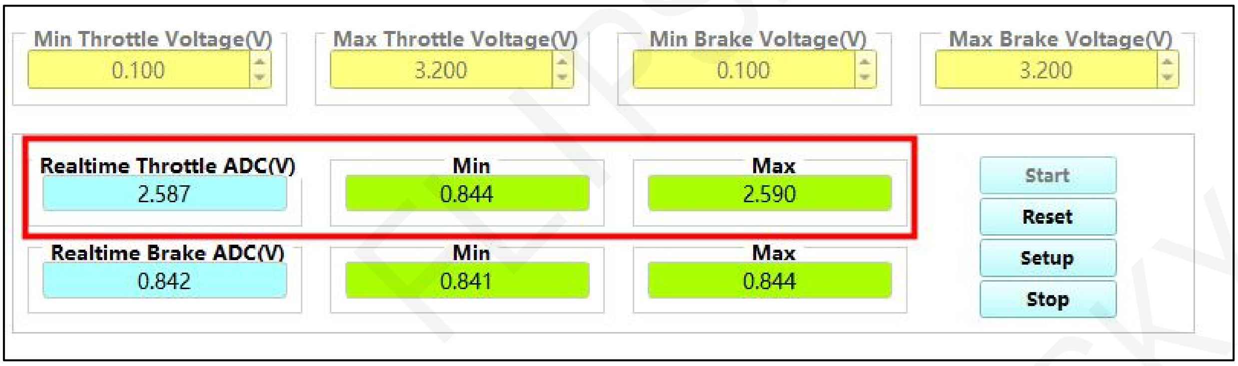
- Click "Start" to begin the calibration as below screenshot 7-24.

Screenshot 7-24 ADC Input Setup



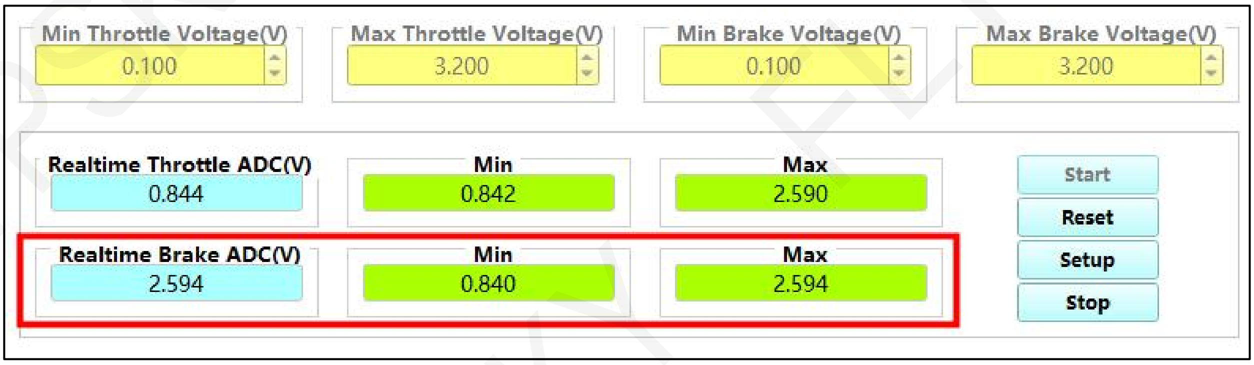
- Max twisting the ADC throttle to flash the min and max voltages as below screenshot 7-25.

Screenshot 7-25 Flash the ADC Throttle Forward



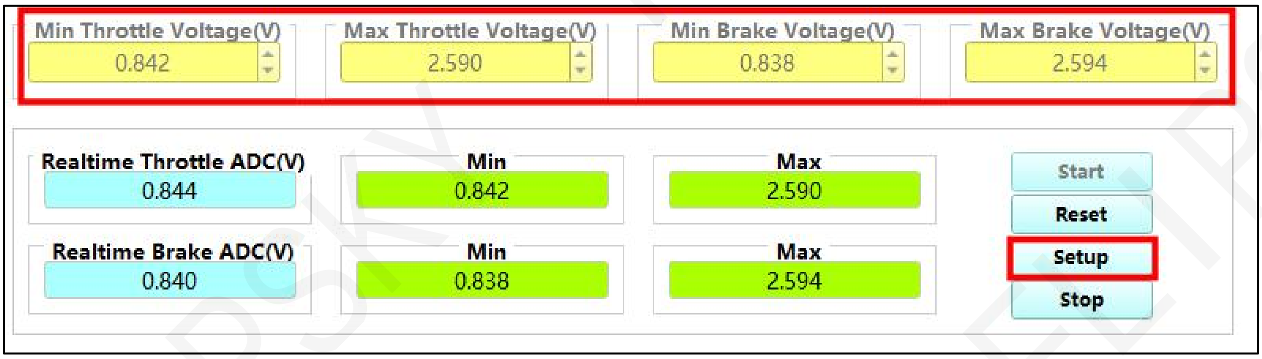
- Max press or pull the brake to flash the min and max voltages as below screenshot 7-26.

Screenshot 7-26 Flash the ADC Brake



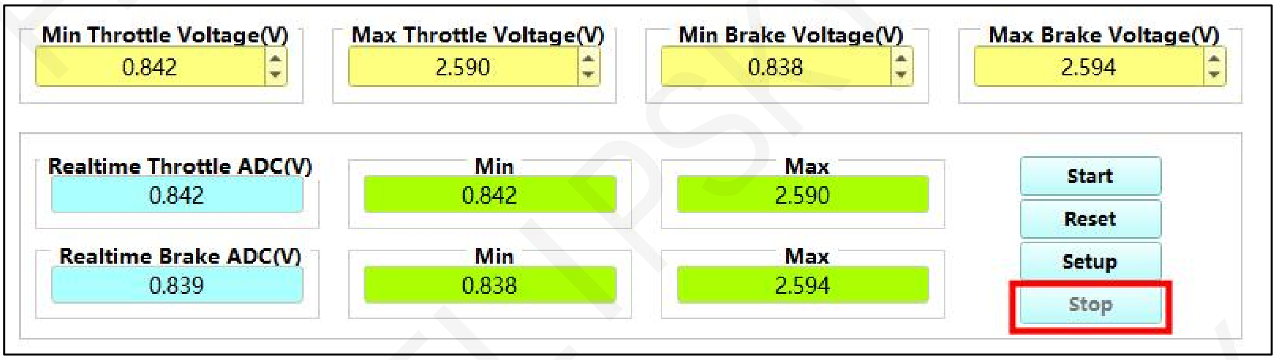
- Click "Setup" to write the "Min" and "Max" data to throttle and brake voltage accordingly as screenshot 7-27.

Screenshot 7-27. ADC Real-time Throttle and Brake Voltage Setup



- Click "Stop" to complete the ADC input setup as screenshot 7-28.

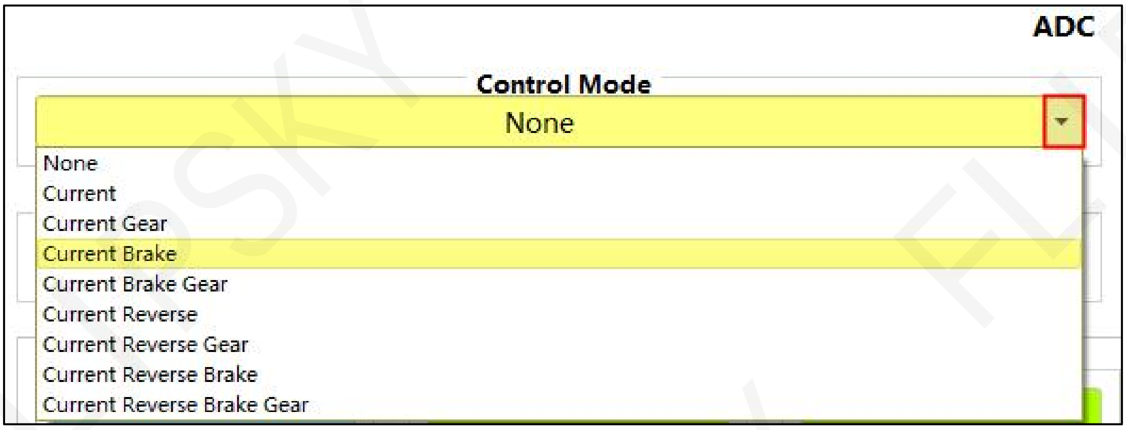
Screen 7-28.ADC Input Setup Complete



7.6.4.3 ADC Control Mode Setup

Select the appropriate ADC Control Mode per your needs as screenshot 7-29.

Screenshot 7-29.ADC Control Mode



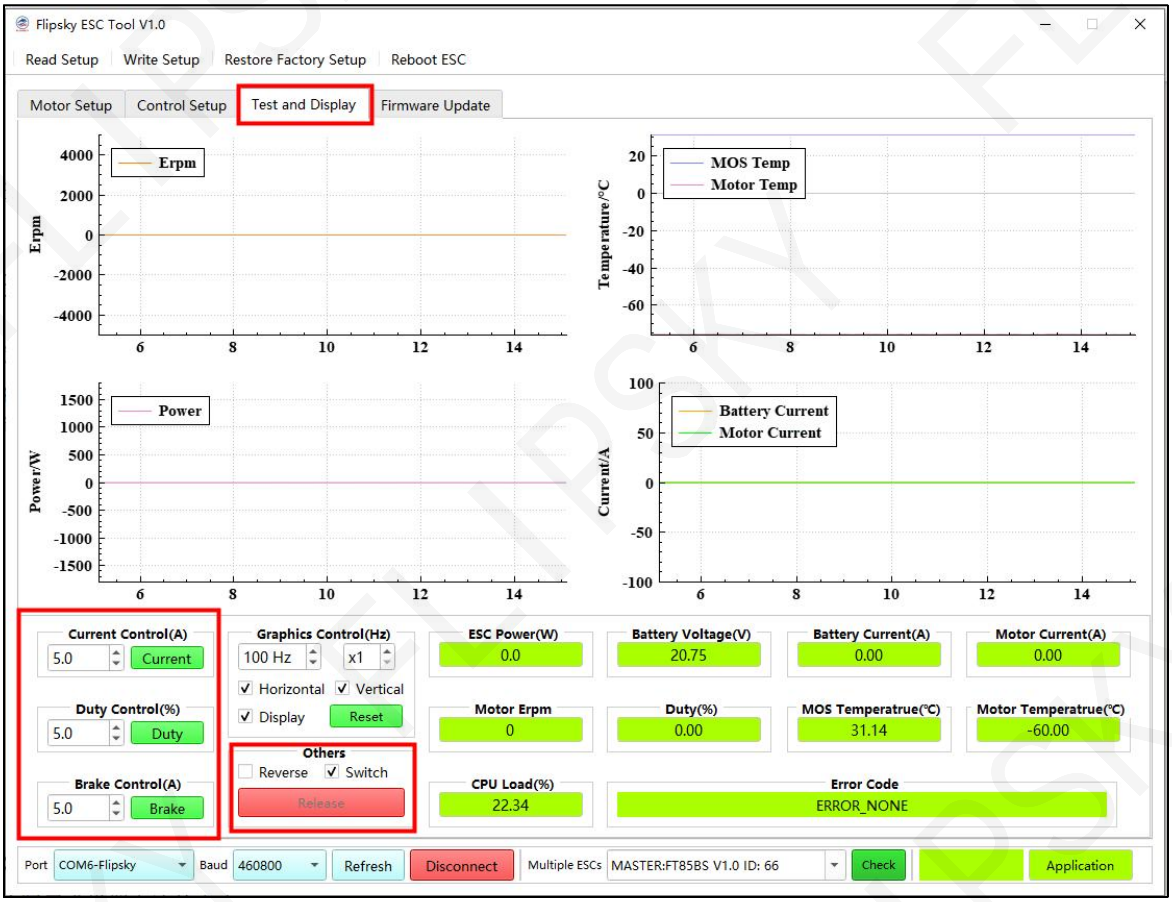
- None: Does not respond to any ADC input signal.
- Current: Current control mode, only enable throttle ADC.
- Current Gear: Current control mode, only enable throttle ADC and L\M\H speed gear.
- Current Brake: Current control mode, enable throttle ADC and brake ADC.
- Current Brake Gear: Current control mode, enable throttle ADC, brake ADC and L\M\H speed gear.
- Current Reverse: Current control mode, enable throttle ADC and reverse gear.
- Current Reverse Gear: Current control mode, enable throttle ADC, reverse gear and L\M\H speed gear.
- Current Reverse Brake: Current control mode, enable throttle ADC, brake ADC and reverse gear.
- Current Reverse Brake Gear: Current control mode, enable throttle ADC, brake ADC, reverse gear and L\M\H speed gear.

8 Test and Display

8.1 Flipsky ESC Tool Button Control

Switch the "Input Signal Type" to "OFF" if you want to control the ESC through the Current button, Duty button or Brake button in "Test and Display" interface as below screenshot 8-1.

Screenshot 8-1 Control Buttons in Test and Display Interface

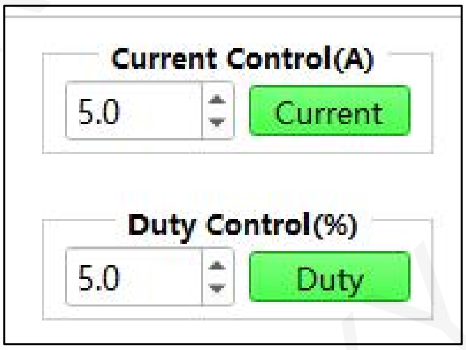


8.1.1 Button Control Setup

How to use the buttons to control the ESC?

- Driving Motors: Click "Current" (Current control mode) or "Duty" (Dutycycle control mode) to drive motors. The current control Amps and dutycycle control ratio can be set per your needs as screenshot 8-2.

Screenshot 8-2. Current and Dutycycle Control Modes

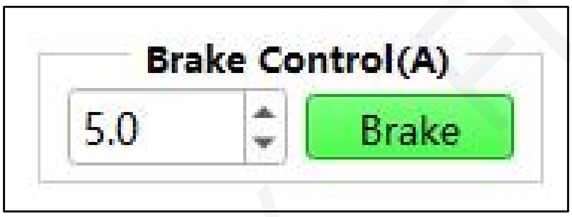


- Motor Rotation Direction Reverse: Tick "Reverse" to change the motor running direction as screenshot 8-3.

Screenshot 8-3 Reverse Direction

- Brake: Click "Brake" to enter the brake setup as screenshot 8-4.

Screenshot 8-4 Brake

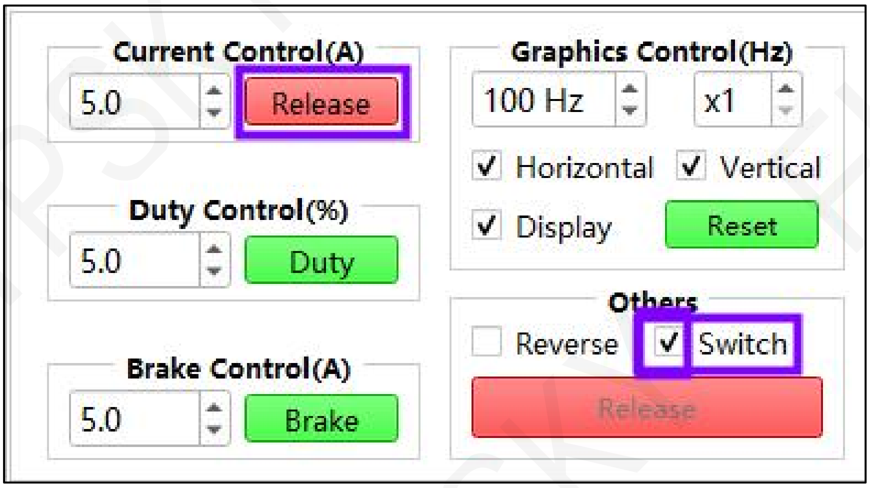


Notes: If using the DC power supply, please do not brake in high speed. Otherwise, the large back EMF will damage the ESC or even the power supply when the motor brakes in a high speed mode.

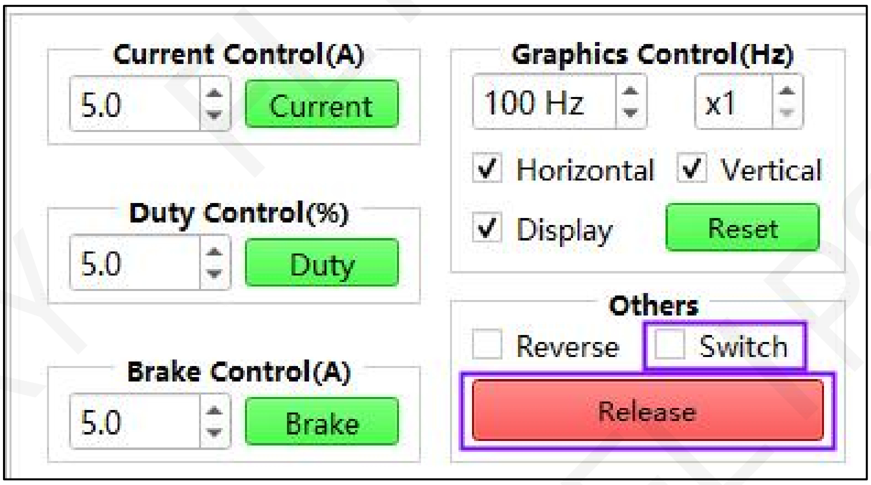
■ Stop: Click "Release" to stop the motor running. Two ways available to stop the motor running: Tick "Switch" and click the"Current" under Current Control, then the "Current" will switch to "Release". Click the "Release" to stop the motor rotating as screenshot 8-5.

If not tick the "Switch" and it’s "Current" under Current Control, the motor can only be stopped through the "Release" under the "Others" as screenshot 8-6.

Screenshot 8-5.Stop Way 1



Screenshot 8-6.Stop Way 2



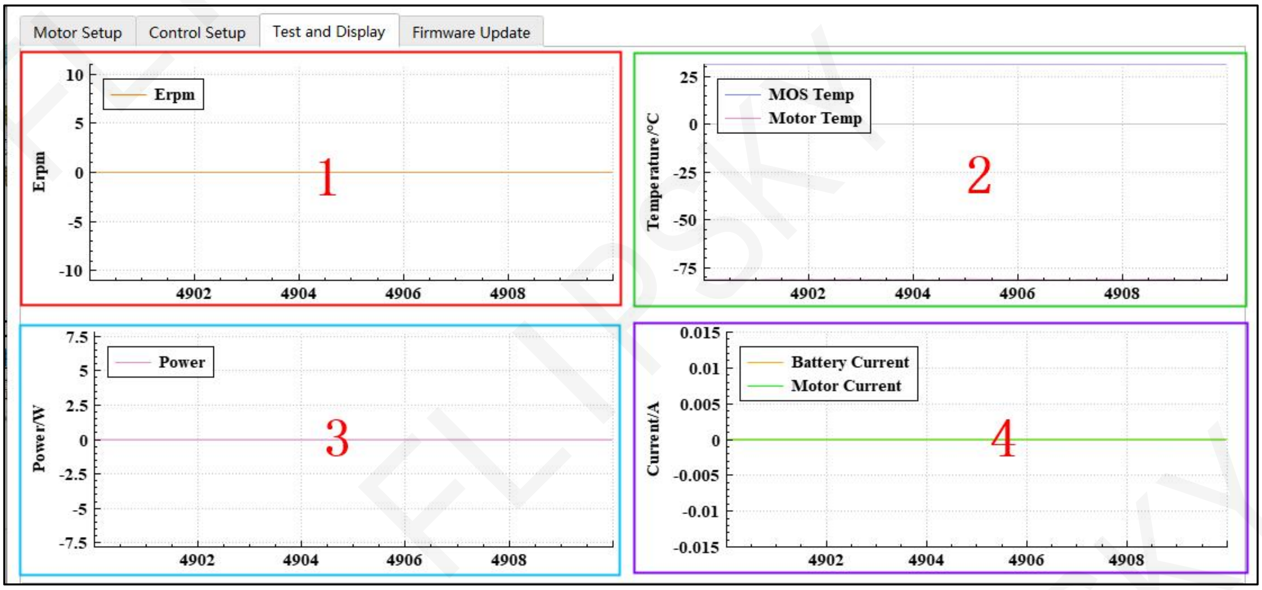
8.2 Data Analysis

8.2.1 Data Curve Display and Setup

There are 4 real-time data graphs as below screenshot 8-7:

- Erpm (Erpm)
- ESC MOSFET and motor temperature curve (Temperature/℃)
- Power curve (Power/W)
- Battery and motor current curve (Current/A)

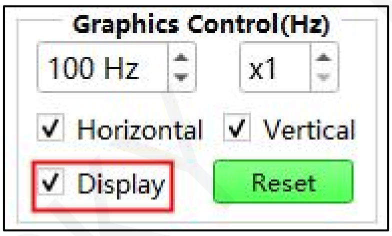
Screenshot 8-7 Real-time data Curve Display Interface



Data Curve Setup Steps:

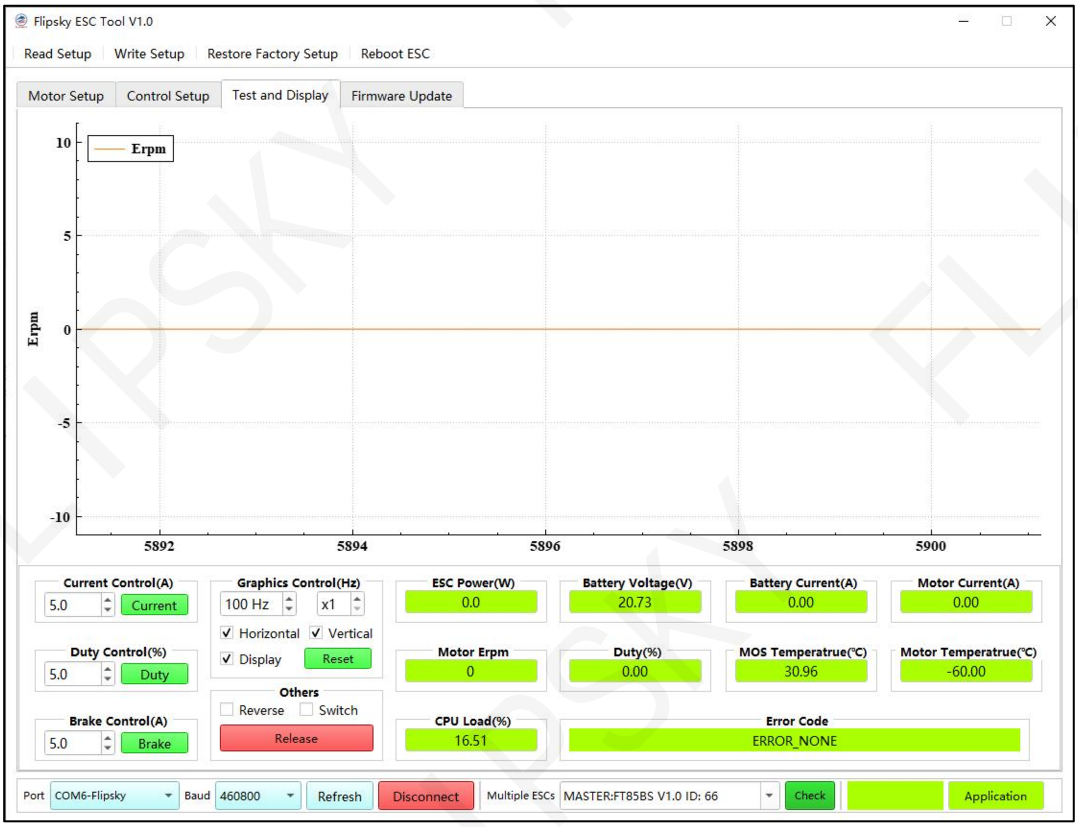
- Activate the graph: Tick "Display" to activate graphics data display as screenshot 8-8.

Screenshot 8-8 Data Display Setup



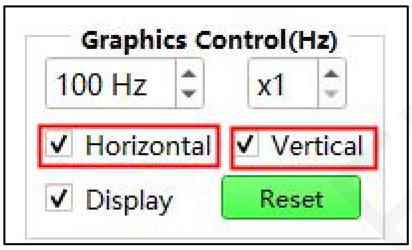
- Graph enlargement: move the mouse cursor in the graph and left-click twice to enlarge it, and restore it in the same way as below screenshot 8-9.

Screenshot 8-9.Graph Enlargement



- Increase or decrease the graph horizontal and vertical axis values: Tick both "Horizontal" and "Vertical" (as screenshot 8-10), move the mouse cursor into the graph and then scroll the mouse wheel to increase or decrease.

Screenshot 8-10 Graphics Control



Notes:

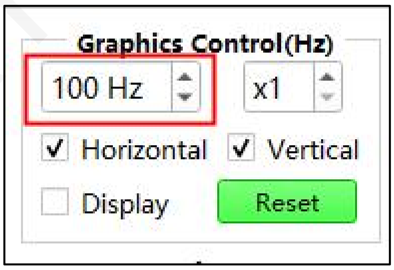
Tick the "Horizontal" and not tick the "Vertical" if you want to increase or decrease the values of horizontal axis only;

Tick the "Vertical" and not tick the "Horizontal" if you want to increase or decrease the values of vertical axis only;

Tick both the "Horizontal" and the "Vertical" if you want to change values of both axis.

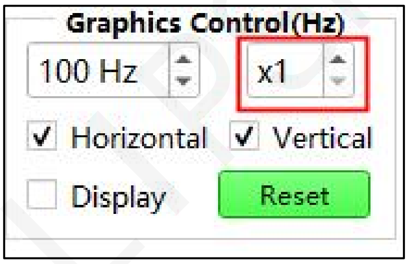
- The sampling frequency setup: default 100Hz, which means that the data samplings are 100 times per second. Adjustable setup range: 50Hz～1000HZ as screenshot 8-11.

Screenshot 8-11 The sampling frequency setup of graphics



- Curve play speed setup: default "X1", adjustable range 1~10 times as screenshot 8-12.

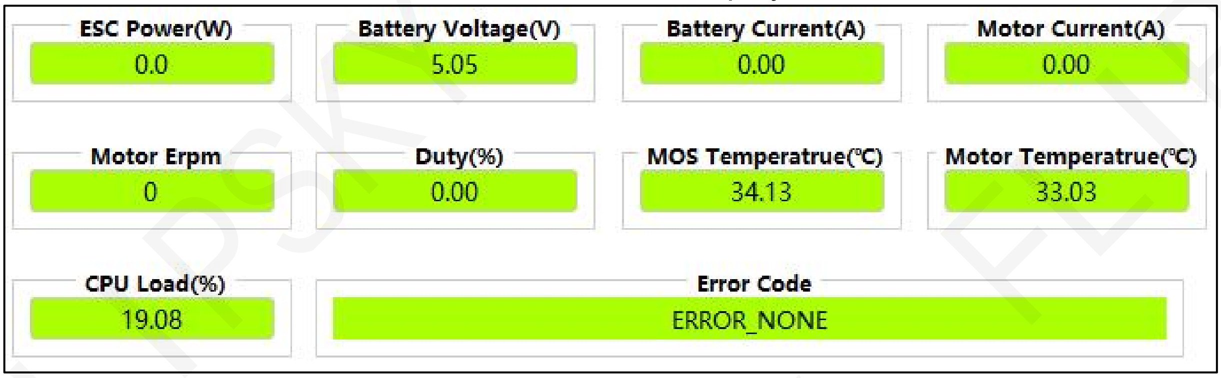
Screenshot 8-12 Curve play speed adjustment



8.2.2 Realtime Data

Data display tables are listed under the graphics curve for better data analysis as screenshot 8-13.

Screenshot 8-13. Data Display Table

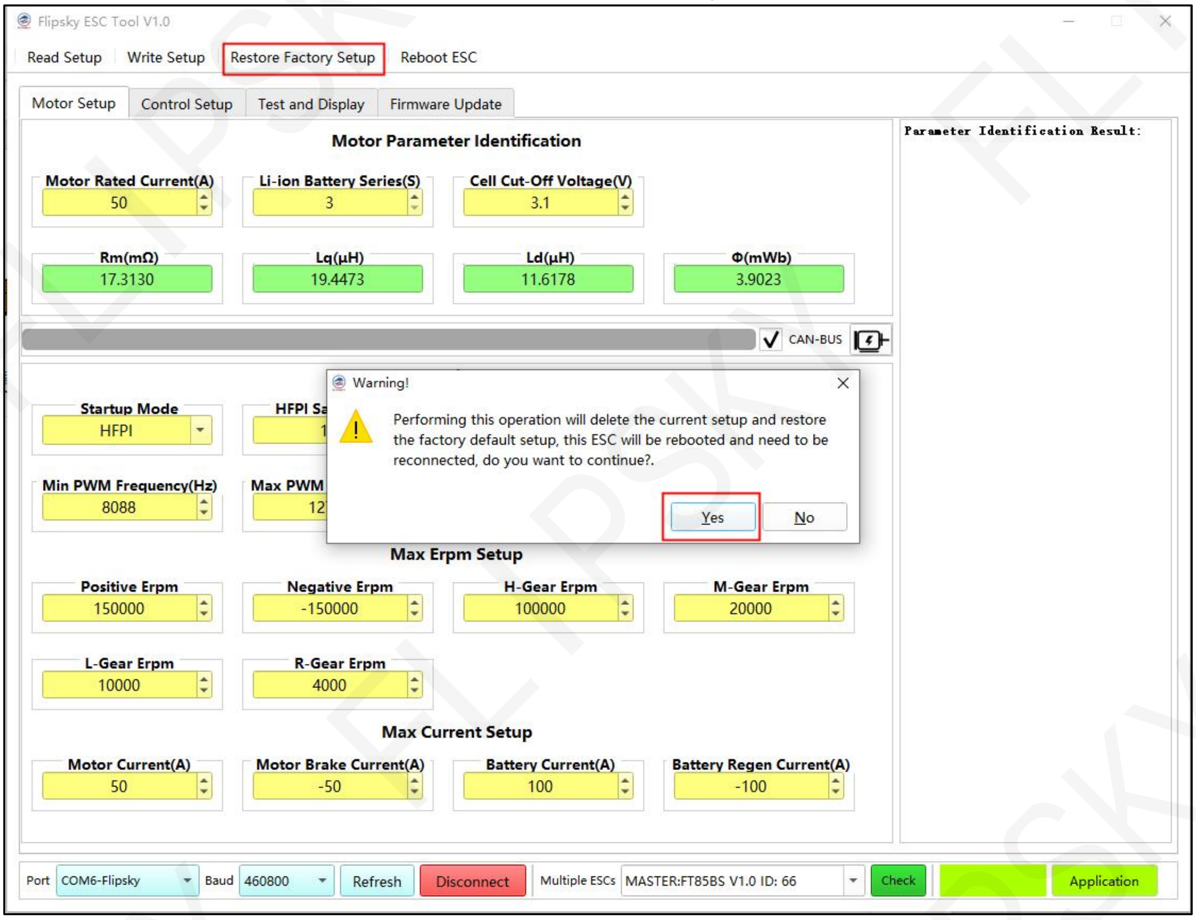


- ESC Power(W): ESC power display
- Battery Voltage(V): battery voltage display
- Battery Current(A): battery current display
- Motor Current(A): motor current display
- Motor Erpm: motor electric rpm display
- Duty(%): Dutycycle percentage display
- MOS Temperature (°C): ESC mosfets temperature display
- Motor Temperature (°C): motor working temperature display
- CPU Load(%): CPU loading percentage display;
- Error Code: fault code display

9 Restore Factory Default Setup

If meeting with some problems like parameters setup failure, ESCs connection failure, motor parameter identification failure etc., please try to restore the factory default setup as screenshot 9-1.

Screenshot 9-1 Restore Factory Default Setup



10 Version

Table 1. File Version

Date	Version	Contents Change
2023.09.06	V1.0	
2023.11.08	V1.1	1. Fixed the problem that the Hall parameter cannot pass when the load is too heavy. 2. Optimize the parameter adjustment procedure. All interfaces are shielded during parameter adjustment to prevent other signals from affecting parameter adjustment. 3. Optimize the dead zone setting mode in ADC mode, and fix the problem that some gamepads may not start smoothly when used; 4. Rectify the fault that the electrical connection to the upper computer fails.